

China AI

China AI: Intelligence is all you need? Initiating coverage on Z.ai and Minimax



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Initiating coverage on China's AI labs. We are initiating coverage on Z.ai on Outperform (PT HK\$1,350), and Minimax on Market-Perform (PT HK\$275). Frontier AI sits at the intersection of a generational tech cycle and global geostrategic competition. Domestically, there is now a clear policy interest in building out a frontier AI industry. Outside China, we expect token rationalisation to help support adoption of open source AI models.

The Pareto frontier, and the right to win. We expect coding to remain the lingua franca between man and machine as agentic AI moves toward recursive self-improvement, and autonomous task completion. Among the AI labs, we expect the winners to be the ones that can (1) deliver differentiated "spikes" in performance; (2) capture convex willingness to pay at the frontier; then (3) serve intelligence cheaply while funding the next training run. For investors, we expect model architecture and performance to remain the core of AI lab analysis, with ARR growth and inference margins representing downstream outcomes.

The next leg of AI adoption. In contrast with AI adoption to date led by 1% of power users spending \$5k monthly, we expect the next leg of AI adoption to be driven by broader-based use, and better matching of token costs with the value of task completion. As more tasks become solvable at scale, we expect the arbiter of competition to shift from reasoning capabilities alone towards cost-effectiveness and availability of compute - we expect the top Chinese AI labs to grow strongly as domestic hardware supply ramps. We estimate a \$100-200bn TAM for AI revenues in China alone, even excluding consumer end use cases.

R&D pedigree is hard to speed-run. While the market caps of these AI lab stocks reflect (often wildly volatile) investor perceptions of their ability to win share in a large AI revenue TAM, the R&D pedigree that powers model architecture innovation and the iteration of frontier reasoning capabilities is hard to change dramatically or quickly. Compared with US peers, we expect the Chinese AI labs to continue to benefit from the lighthouse effect of frontier R&D. Meanwhile, we think the magnitude of the scientific problem at hand means the frontier is more likely to become less crowded over time.

Z.ai: the K3 reset was needed, but the future remains bright. Kimi K3 puts the onus on Z.ai to deliver on its next model, which we expect by Q4 (after a GLM-5.3 update this month). In the meantime, GLM-5.2 showed impressive intelligence for a 744bn parameter pre-train, remains on the Pareto frontier for cost-effectiveness, and we expect Z.ai's revenue to exceed consensus estimates in the coming years. Even on assumptions of rapid R&D spend growth, we think Z.ai can achieve non-GAAP breakeven circa 2028.

Minimax: a lot rides on M3 Pro. Minimax has a history of strategic pivots in recent years, and the M3 disappointment makes M3 Pro feel rather make-or-break. Recapturing frontier credibility would of course be game-changing for Minimax's shares, but going from 428bn to 2.7tn total parameters would be more than double the "typical" step-up among Chinese peers. H3 should give ARR growth a boost, but AI video generation arguably has a lower ceiling for commoditisation and revenue TAM than agentic coding.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	4 Aug 2026	Price	TTM	Adjusted EPS				Adjusted P/E (x)		
			Closing	Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
2513.HK (Z.AI)	O	HKD	1,014.00	1,350.00	NA	RMB	(2.40)	(7.95)	(5.26)	(423.1)	(127.5)	(192.7)
100.HK (MiniMax)	M	HKD	230.20	275.00	NA	USD	(2.31)	(0.97)	(1.36)	(12.7)	(30.3)	(21.5)
ASIAX			1,897.76									

COVERAGE INITIATION

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We're initiating coverage of Z.ai on an Outperform rating (PT HK\$1,350) and Minimax on a Market-Perform rating (PT HK\$275). This note consolidates some of our YTD work on the state of China AI, and outlines some top-down views on the broader peer group as more of China's top AI labs go public.

We're bullish at a high level on the top Chinese AI labs' ability to stay close to the global AI frontier, and expect them to grow rapidly as improving reasoning capabilities unlock new use cases, and as domestic compute constraints ease over time. The top Chinese AI labs benefit from the lighthouse effect from frontier R&D. Domestically, their success is likely a matter of national strategic importance. Outside China, we're hopeful token rationalisation and a greater focus on cost-effectiveness will support adoption of Chinese open source models.

Bottom up, we expect model capabilities to remain the chief driver of AI lab share price performance. The reset in Z.ai's share price post Kimi K3 felt representative of the type of reset in long-term expectations that can take place in a volatile space. But the company is led by foundational scientists with impressive R&D pedigree, GLM-5.2 has shown impressive intelligence for its total parameter scale, and the company has expressed confidence that the next GLM pre-train will be China frontier. Financially, 1GW of available compute by year-end should mean Z.ai's ARR can exceed \$2bn either side of year-end. We model non-GAAP operating profit breakeven by 2028, and value the company on a combination of 25x 2030E PE discounted back, and our DCF for the company.

Minimax's shares have fallen a long way from peak levels, and we get the desire to look on the bright side. But the M3 disappointment means M3 Pro is now make-or-break, requiring a jump from 428bn to 2.7tn total parameters - more than double what has been "typical" among peers. Minimax faces less competition in the multimodal market, and we expect the H3 model to help support ARR growth. But AI video generation likely has a lower ceiling for commoditisation and overall revenue TAM than agentic coding and productivity work. We value the company on a blend of 4.5x 2029E PS and our DCF for the company, but expect to revise this in response to the M3 Pro model outcome.

EXHIBIT 1: **Our ratings for Z.ai and Minimax focus chiefly on R&D pedigree and AI model progress**

	Z.ai 2513.HK	Minimax 100.HK
Rating	OUTPERFORM	MARKET-PERFORM
Current price	HK\$ 940	HK\$ 247
Price target	HK\$ 1,350	HK\$ 275
Upside/downside potential	43.6%	11.2%
Reporting currency	RMB	USD
Consensus ratings	21 	25 
Market cap (US\$ bn)	55	10
Enterprise value (US\$ bn)	53	8
Revenue	RMB mn	US\$ mn
2026E	5,631	419
2027E	18,663	1,005
2028E	44,435	2,217
2026E-2028E CAGR	180.9%	130.0%
2026E vs. consensus	+35.4%	+16.5%
2027E vs. consensus	+52.6%	+2.1%
2028E vs. consensus	+61.9%	+4.8%
EPS	RMB	US\$
2026E	-7.95	-0.97
2027E	-5.26	-1.36
2028E	3.84	-0.75
2026E-2028E CAGR	n.a.	n.a.
2026E vs. consensus	+29.3%	+47.2%
2027E vs. consensus	+50.9%	+27.2%
2028E vs. consensus	+184.5%	+47.9%
P/E		
2026E	n.a.	n.a.
2027E	n.a.	n.a.
2028E	211.1x	n.a.
EV/sales		
2026E	63.1x	18.2x
2027E	19.0x	7.6x
2028E	8.0x	3.4x

Pricing as of Aug 3, 2026.

Source: Corporate reports, Bernstein estimates and analysis.

VALUATION COMPS TABLE

EXHIBIT 2: Asia Internet & AI: valuation summary

	Rating	Price	Last	Crncy	Market cap (US\$mnn)	PE			EV/sales		
		target	price			2026E	2027E	2028E	2026E	2027E	2028E
China Internet coverage											
Tencent	O	780	490.40	HKD	568,594	14.2x	12.2x	10.6x	4.7x	4.1x	3.6x
PDD	M	110	88.56	USD	126,056	8.2x	7.3x	6.5x	0.8x	0.5x	0.2x
Meituan	M	85	93.45	HKD	73,581	209.3x	19.4x	11.3x	1.0x	0.9x	0.7x
NetEase	O	150	133.27	USD	85,043	14.7x	13.6x	12.9x	3.6x	3.3x	3.1x
Boss Zhipin	O	18	16.28	USD	7,707	13.2x	10.9x	9.8x	3.5x	2.5x	2.0x
JD	O	40	33.01	USD	45,086	9.4x	7.1x	5.4x	0.2x	0.2x	0.2x
Alibaba	O	180	122.25	USD	293,041	30.6x	18.5x	15.2x	2.0x	1.9x	1.7x
Z.ai	O	1,350	940.00	HKD	55,812	-160.5x	8364.1x	76.5x	63.2x	19.3x	8.1x
Minimax	M	275	247.20	HKD	11,009	-39.0x	-32.4x	-114.8x	19.9x	8.8x	3.8x
China Internet other											
Kuaishou			45.02	HKD	24,839	10.5x	9.2x	8.0x	1.0x	0.9x	0.9x
Bilibili			19.10	USD	8,118	18.3x	14.8x	11.4x	1.2x	1.1x	1.0x
TME			9.45	USD	15,702	10.0x	9.1x	8.3x	2.1x	1.9x	1.8x
Baidu			111.11	USD	37,805	16.0x	13.3x	10.7x	1.1x	1.0x	0.9x
VIPshop			15.34	USD	7,367	6.1x	5.8x	5.5x	0.3x	0.3x	0.3x
Tencent Music			9.45	USD	15,702	10.0x	9.1x	8.3x	2.1x	1.9x	1.8x
Trip.com			47.01	USD	29,602	13.7x	11.7x	10.5x	2.3x	2.0x	1.8x
KE Holdings			16.95	USD	19,616	18.3x	15.6x	15.5x	1.2x	1.1x	1.1x
Asian Internet											
Naver	O	330,000	204,000	KRW	22,422	14.5x	12.9x	11.3x	1.5x	1.2x	0.9x
Kakao	O	80,000	37,200	KRW	11,541	20.9x	18.5x	15.2x	1.1x	0.8x	0.6x
Hybe	O	400,000	172,000	KRW	5,193	16.9x	16.8x	15.2x	1.0x	1.2x	1.0x
Coupang	U	12	16.35	USD	29,349	-43.1x	77.6x	37.7x	0.7x	0.6x	0.6x
Sea Ltd.			106.74	USD	65,377	28.7x	21.2x	16.6x	1.9x	1.6x	1.4x
Grab			3.50	USD	14,352	35.0x	23.0x	17.1x	2.4x	2.0x	1.7x
US Internet											
Amazon			271.58	USD	2,929,347	26.5x	22.3x	17.4x	3.7x	3.2x	2.9x
Alphabet			356.13	USD	4,358,561	18.6x	21.5x	18.2x	9.8x	7.8x	6.4x
Meta			556.71	USD	1,418,222	15.6x	14.1x	11.7x	5.7x	4.7x	4.0x
Netflix			71.71	USD	298,596	20.2x	18.7x	15.7x	6.0x	5.4x	4.9x
Uber			70.36	USD	143,225	21.3x	15.8x	12.8x	2.6x	2.3x	2.0x
Spotify			499.94	USD	102,904	34.7x	27.5x	22.3x	4.2x	3.6x	3.2x
DoorDash			196.16	USD	85,471	36.2x	29.5x	22.1x	4.7x	3.9x	3.3x

Pricing as of Aug 3, 2026.

Source: Corporate reports, Bloomberg, Bernstein estimates and analysis.

DETAILS

INITIATING COVERAGE ON CHINA'S AI LABS - Z.AI OUTPERFORM, MINIMAX MARKET-PERFORM

We are initiating coverage of Z.ai with an Outperform rating (PT HK\$1,350) and Minimax with a Market-Perform rating (PT HK\$275). We consider the former a cleaner expression of repeatable frontier-reasoning R&D capability, while an M3 Pro success now feels make-or-break for the latter... requiring a much-larger-than-typical jump in total parameter scale versus the current M3 model. We also feel that multimodal has lower ceilings for commoditisation and revenue TAM compared to coding and agentic AI.

Taking a step back, it seems beyond doubt that AI reasoning capabilities will continue to iterate rapidly. Autonomous task completion and recursive self-improvement have become consensus focus areas for next-generation AI model capabilities (see Exhibit 3). The evolution of US and Chinese frontier model capabilities will remain both scientific and geo-strategic focal points in the coming years (see Exhibit 4 and Exhibit 5). We expect the top Chinese AI labs to continue to show progress, helped by elite research pedigree, and over time a growing and improving supply of domestic silicon. Our base case for the shape of AI adoption is demand will increasingly bifurcate across frontier labs solving a smaller number of more difficult, higher-ARPU tasks... and the need for token rationalisation behind the frontier, where token costs are matched with the incremental value of task completion. Domestically, the likes of Z.ai, Kimi, and Qwen will compete to be the frontier. Internationally, we expect them to grow by serving “good enough” reasoning capabilities at much lower prices.

In contrast with the “user acquisition, engagement growth, and monetisation of eyeballs” playbook that drives our Internet platform coverage, we think the path to commercial success for the AI labs centres around delivering frontier capability “spikes” that attract a core group of users willing to pay... which then supports credibility and fundraising that allows them to make the next technology leap. A place on the Pareto frontier for reasoning capabilities versus price will be table stakes - it feels at least a little notable that the current Pareto frontier consists of Claude Opus 5... and a collection of Chinese models (see Exhibit 6). In a future where AI models are orchestrated to their strengths, we expect the labs’ ability to offer notable capability spikes to be at least as important as showing well-rounded strength.

For AI users and developers, superior task completion drives repeat usage and deeper workflow embed. An emerging cohort of agentic harness apps have the potential to help improve real-world model usefulness (by adding persistent memory, a library of skills and reference files, etc.), and build usage stickiness. For the AI lab, adoption translates into developer feedback, distribution support, and revenue and credibility which contribute to the next training run. The reverse can also be true: a poor model can quickly drive away users, impacting both revenue momentum and the cost of funding for the next step forward.

Research pedigree is a hard thing to speed-run

For all the discussion around AI model commoditisation, training frontier reasoning capabilities remains a difficult scientific endeavour. That said, the market’s ability to be surprised by Chinese AI progress has struck us as curious for some time, given the battle for global AI supremacy now seems to be fought between two groups of engineers with similar educational and research backgrounds... who happen to work for competing AI labs. It goes without saying that the US labs will continue to enjoy significantly better access to compute. On the other hand, there’s been a long history of China’s top AI labs recruiting from frontier labs to bolster their R&D capabilities, and to date the Chinese AI labs have done a good job offsetting compute shortfalls with architectural innovations.

We consider AI model iteration a capability-driven rather than probability-based process. In contrast with common investor perceptions of AI labs to date, which have tended to be very headline driven - often anchored on the *last* published model, R&D pedigree is both critical and hard to change dramatically over short time periods. Earlier this year, Minimax and Z.ai were initially considered in a similar light, until the GLM-5 and M3 launches pointed to clearly differentiated pre-train scale and reasoning capabilities. The Kimi K3 launch set a new bar for pre-train scale, and drove a large drop in Z.ai’s shares (see Exhibit 7). We expect the merry-go-round of investor perceptions to continue, but expect this to continue to drive trading opportunities across a volatile group of stocks.

We think the magnitude of the scientific problem that frontier AI labs are trying to solve means the frontier is more likely to become less, rather than more, crowded over time. The large Internet platforms we cover benefit from significant scale and funding advantages, and the recent improvement in Tencent’s AI progress perhaps reflects that AI development muscle can be bought to some degree. Bytedance’s ability to throw more money at the problem will remain relevant too, despite recent issues. Qwen remains frontier, even if its large models have yet to reach the same level of community acclaim as the Qwen small models. On the other hand, more often than not, our discussions with AI researchers in the field have pointed to a preference for

working in environments where chasing frontier AI development is the singular focus, and R&D velocity is faster - as opposed to “building a better toaster” and dealing with bigco bureaucracy and politics.

While we’ve generally been impressed by the R&D pedigree at Z.ai, staffed by foundational scientists, it feels notable that the backgrounds of Minimax leadership have historically skewed towards computer vision and traditional natural language processing, and did not follow the more “typical” transformer architectures/LLMs research path that defines most of China’s other top AI lab leaders (see Exhibit 8). Z.ai counts most of China’s top Internet platforms among its customers, and its Tsinghua University affiliation likely contributes to funnels for data and talent. While Z.ai, Kimi, and DeepSeek have generally seen greater support from global TaaS platforms over time, Minimax notably has not (see Exhibit 9). Kimi’s K3 commercial licence (requiring licencing over a low hurdle for revenue, and seemingly requiring TaaS partners to avoid undercutting the first-party API) sets an interesting precedent which we think will help the AI labs capture more value from global adoption.

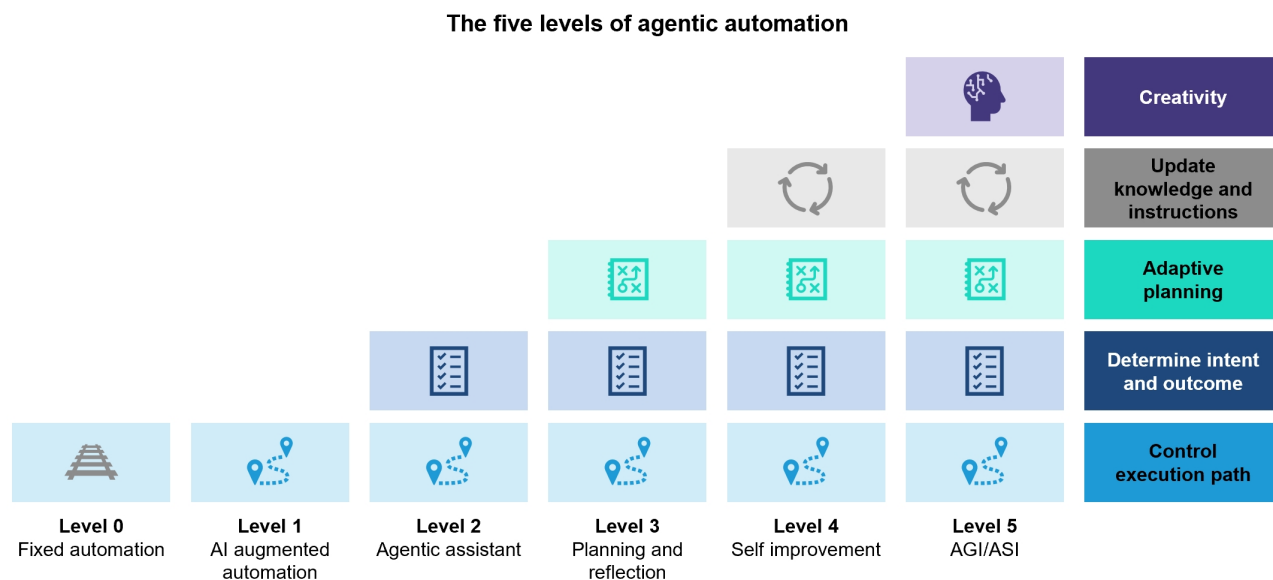
We rate Z.ai Outperform (PT HK\$1,350), Minimax Market-Perform (PT HK\$275)

Given the pace of technology evolution at the AI frontier, we fully expect the recent volatility in Z.ai and Minimax’s shares to continue. The scarcity premium that was ascribed to Z.ai shares earlier this year was erased in recent months as the free float of the stock became larger, and as Kimi K3 came to market. That said, given the size of the long-term opportunity, and the importance of frontier reasoning capabilities to China’s geostrategic objectives, we expect the long-term payoff for China’s AI leaders to be positive.

Z.ai top leadership has established frontier AI research pedigree, and we expect the company’s focus on coding to remain well aligned with the nature of enterprise AI adoption ongoing both in China and globally. Z.ai’s GLM-5.2 release was a breakthrough moment for the company, and our high-frequency data pointed to a sharp spike in web page visitation as well as API revenue. The Kimi K3 release puts the onus on Z.ai to deliver on the next step-up in total parameter scale. But we think snap reactions among investors post K3, arguing that Z.ai was “no longer frontier” were premature. The company has pointed to a GLM-5.3 update this month, with the next pre-train expected to follow in Q4. Our price target for Z.ai is driven by a blend of a 25x 2030E PE multiple, discounted back at 14% annually, and our DCF for the company.

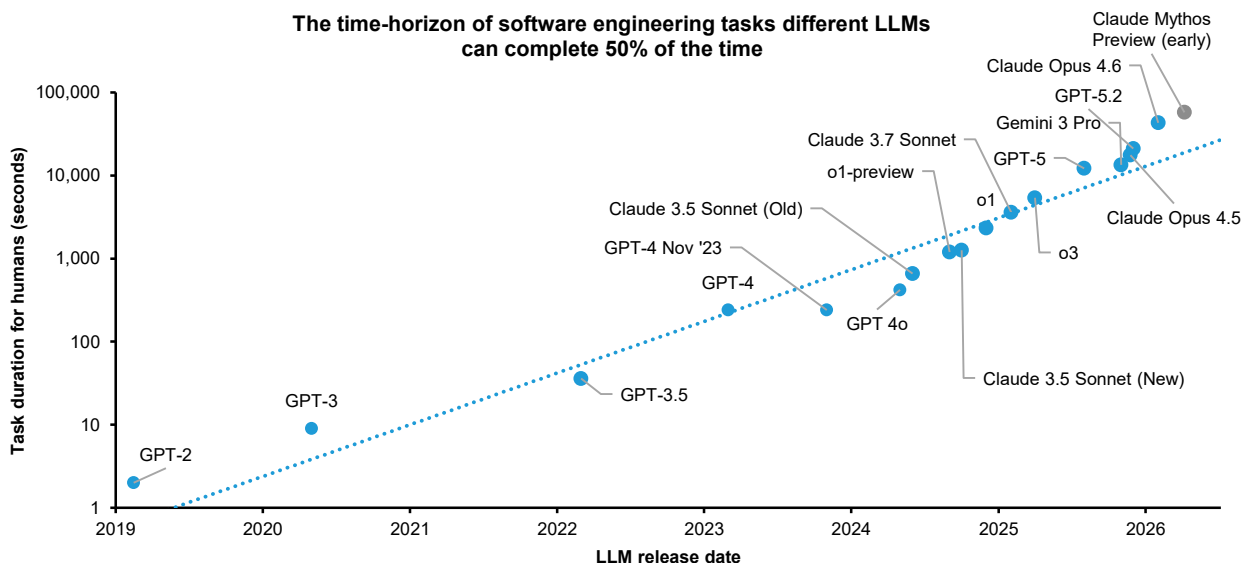
We expect Minimax ARR to grow too. But at this point M3 has fallen a long way behind the frontier, making M3 Pro a make-or-break moment - requiring a much-larger-than-typical step-up in parameter scale from its last published model, compared with peers in recent years. In the meantime, DeepSeek V4-Flash 0731 underscored the potential for price competition at the lower-complexity end of the market. The H3 release was encouraging, and likely helps to reset positioning to some degree. But we remain of the view that multi-modal generation has a lower ceiling for commoditisation, and therefore deserves lower terminal multiples. Our price target for Minimax is based on a blend of 4.5x 2029E PS, discounted back at 14% annually, to reflect the relative lack of model pipeline clarity, as well as our DCF for the company.

EXHIBIT 3: The world is spinning too fast... autonomous task completion and recursive self improvement have become the newest consensus focal points for AI development



Source: Sema4.ai and Bernstein analysis.

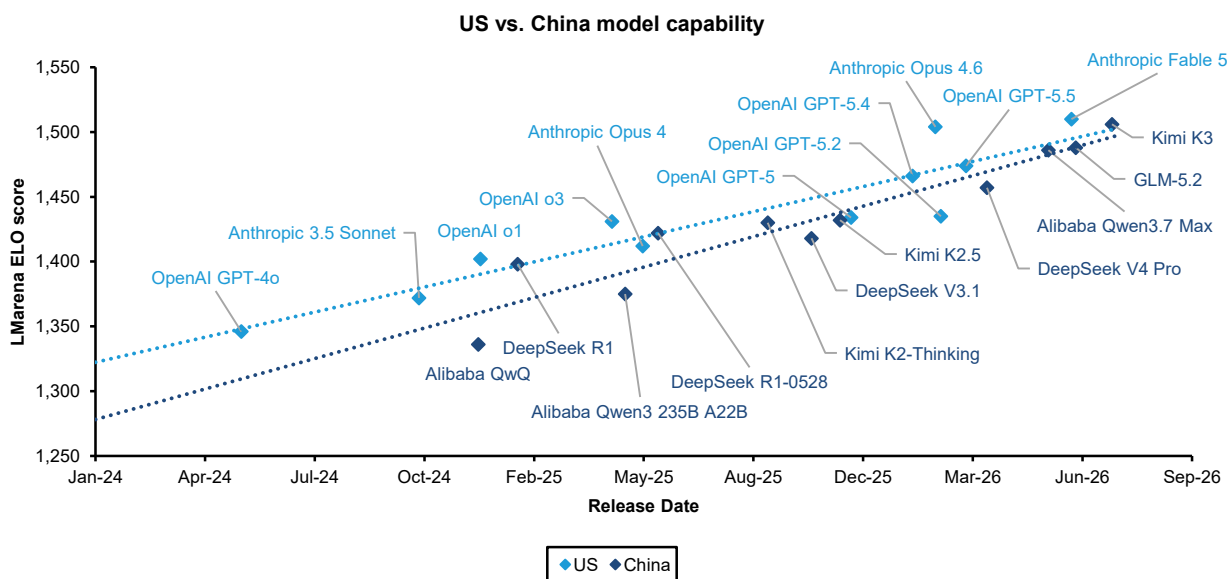
EXHIBIT 4: We expect the ability of agentic AI to solve tasks over longer horizons to continue to show significant progress in the coming years, and support the displacement of parts of human white collar labour



METR thinks measurements above 16 hrs are unreliable with their current task suite.

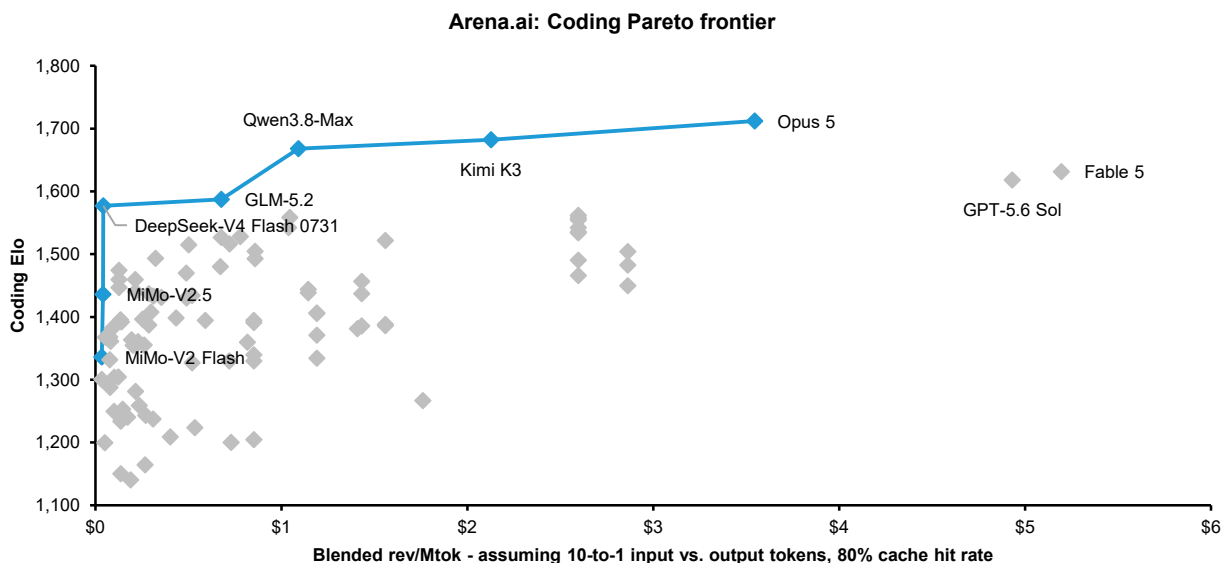
Source: METR and Bernstein analysis.

EXHIBIT 5: Our base case is the US frontier labs remain ahead on frontier intelligence, while the top Chinese AI labs continue to keep pace with a lag measured in months



Source: LMarena.ai and Bernstein analysis.

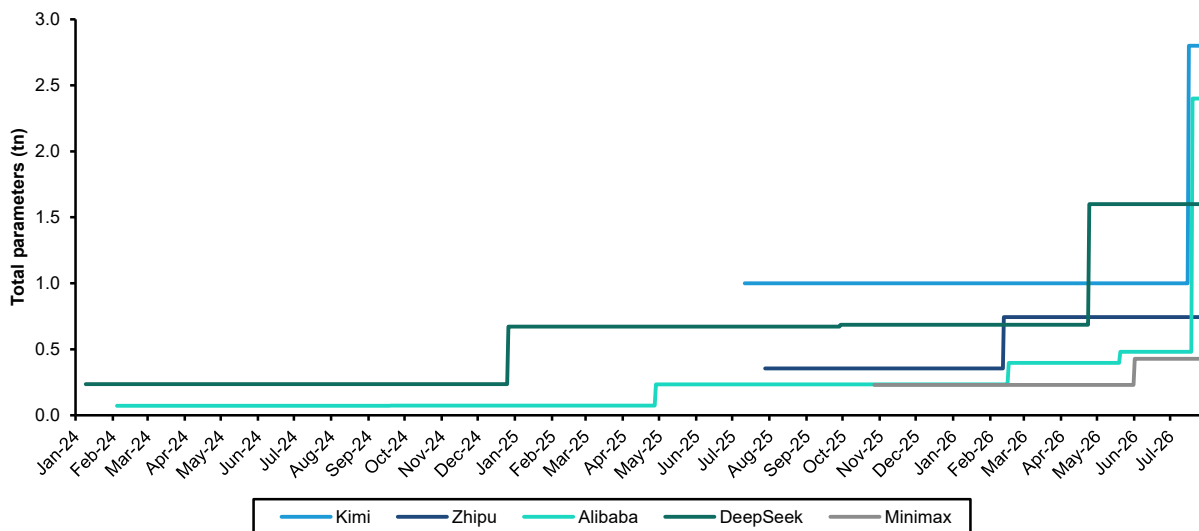
EXHIBIT 6: For individual AI labs, having a model on the Pareto frontier will be table stakes... we expect “spikes” in performance to be at least as important as well-rounded model output



Source: Arena.ai and Bernstein analysis.

EXHIBIT 7: Investor views on these labs have been very volatile, and often backward looking... which likely continues to give rise to trading opportunities, especially as more of China's AI labs come to market

2024-2026: Chinese AI model total parameter scale



Source: Corporate reports and Bernstein analysis.

EXHIBIT 8: Model iteration progress reflects R&D pedigree... we've referenced Google Scholar citations as a somewhat quantitative measure of the relative standings of the leading China AI labs

Lab	Researcher	Role	Research focus	Prior Institutions	Citations - all	Citations (2021-)	h-index	i10-index
Z.ai	Jie Tang	Co-founder	Knowledge graphs, social networks, LLMs	Tsinghua; AMiner founder	79,388	61,682	126	461
	Aohan Zeng	Lead researcher	LLM, instruction tuning, agents	Tsinghua	10,227	10,214	19	23
	Xin Lv	Reasoning RL lead	Reasoning RL, RL Infra, Agentic RL	Tsinghua	7,408	7,228	25	33
MiniMax	Junjie Yan	Founder/CEO	CV, object tracking, deep learning	SenseTime	37,364	27,312	84	127
	Pengyu Zhao	LLM research lead	Streaming media, recommendation algorithms	Hulu	n.a.	n.a.	n.a.	n.a.
Moonshot	Zhilin Yang	Co-founder/CEO	LLM pre-training, long context, QA	Google Brain; CMU; Tsinghua	52,100	44,661	42	59
	Yuxin Wu	Co-founder	ML systems, pre-training, CV	Google Brain; Tsinghua	44,217	40,360	22	22
	Xinyu Zhou	Co-founder	LLM systems and infrastructure, CV	Megvii; Tsinghua	25,684	21,494	24	27
DeepSeek	Wenfeng Liang	Founder/CEO	Quantitative finance, ML	Zhejiang U; High-Flyer	n.a.	n.a.	n.a.	n.a.
	Yu Wu	LLM alignment lead	LLM alignment, RLHF, reasoning	Beihang U	52,802	51,524	53	90
	Wen Liu	LLM architect	LLM architecture, multimodal	Tencent	32,002	31,455	31	34
Alibaba Qwen	Steven Hoi	Head of multimodal interaction	Multimodal models, agents, ML	SMU; Salesforce AI; NTU	81,983	68,206	104	335
	Baosong Yang	Senior algorithm expert	Machine translation, multilingual LLM	University of Macau	22,059	21,677	37	69
	Hao Zhou	Head of post-training research	Post-training, alignment, evaluation	Google DeepMind	10,653	9,989	44	83
Bytedance Seed	Yonghui Wu	Head of foundational research	LLM pre-training, ASR, Seq2Seq	Google	85,112	72,203	85	152
	Bowen Yu	Post-training lead	Post-training, RLHF, instruction tuning	Alibaba; CAS	48,153	48,074	54	106
	Hang Li	Director of AI lab	NLP, information retrieval, ML	Microsoft Research Asia; Huawei	39,498	18,245	88	242
	Chengyi Wang	Audio/speech lead	Speech synthesis, audio LLM, TTS	Microsoft Research Asia; Nankai U	11,460	11,412	25	30
Tencent Hunyuan	Vinces Yao	Chief AI scientist	Language agents	OpenAI; Princeton	39,567	39,469	29	35
	Yonglong Tian	Head of multimodal	Multimodal representation learning, CV	OpenAI; MIT	29,737	26,896	36	45
	Tianyu Pang	Tech lead of multimodal reinforcement learning	Multimodal reinforcement learning	Sea AI Lab; Tsinghua	19,294	17,770	52	96
Xiaomi	Fuli Luo	MiMo team lead	NLP, LLM architecture, pre-DeepSeek; PKU training		26,473	26,199	32	41

Source: Google Scholar, corporate reports, and Bernstein analysis.

EXHIBIT 9: The growth in overseas partners GLM-5.2 and Kimi K2.6 attracted reflects our view that intelligence will continue to be the primary variable that drives go-to-market... Minimax has notably seen a drop in distribution partners

Provider	Z.ai			Minimax			Kimi			DeepSeek		
	GLM 5	GLM 5.1	GLM 5.2	M2.5	M2.7	M3	K2.5	K2.6	Kimi K3	V3.1	V4 Pro	V4 Flash
Official	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Novita	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓
AtlasCloud	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓
DeepInfra	✓	✓	✓		✓	✓	✓	✓		✓	✓	✓
SiliconFlow	✓	✓	✓	✓			✓	✓		✓	✓	✓
Venice	✓	✓	✓	✓		✓	✓	✓			✓	✓
DigitalOcean	✓	✓	✓	✓			✓	✓	✓		✓	✓
Parasail	✓	✓	✓			✓		✓			✓	✓
StreamLake	✓	✓	✓	✓			✓	✓			✓	✓
CoreWeave		✓	✓	✓				✓		✓	✓	✓
GMICloud	✓	✓	✓		✓	✓					✓	✓
Fireworks		✓	✓		✓			✓	✓		✓	✓
Baidu	✓	✓	✓					✓			✓	✓
Chutes	✓	✓	✓	✓			✓	✓				
Phala	✓	✓	✓	✓			✓	✓				
Together			✓		✓	✓		✓	✓		✓	
BaseTen			✓					✓	✓		✓	
Alibaba			✓								✓	✓
Cloudflare			✓					✓				✓
Friendli		✓	✓	✓				✓				
Inception			✓	✓				✓				
Ionstream			✓								✓	✓
Mara				✓	✓					✓		
Morph			✓			✓			✓			✓
Wafer		✓	✓						✓		✓	
AkashML			✓									✓
Ambient			✓									✓
Crusoe		✓						✓				
Decart			✓					✓				
Io Net			✓									✓
SambaNova					✓					✓		
Amazon Bedrock	✓									✓		
Google										✓		
Groq					✓							
Mancer 2												✓
Modal									✓			
ModelRun								✓	✓			
Nebius		✓										
Sail Research			✓									
Total external providers	13	18	28	13	9	8	9	20	7	8	17	20

Source: Openrouter.ai and Bernstein analysis.

Intelligence drives commercial convexity at the frontier, but...

We expect AI lab growth to continue to be non-linear in nature, swinging between (1) R&D and model capability breakthroughs driving periods of higher developer adoption, commercial convexity at the frontier, better distribution support from compute partners, and jumps in revenue... and (2) the arbiter of competition slowly shifting back towards cost-effectiveness, reliability, and distribution as tasks become solvable by multiple AI models.

For most of recent history, the top Chinese AI models maintained a 6-9 month gap with US SOTA - though GLM-5.2 and Kimi K3 recently may argue for a smaller one (see Exhibit 10). DeepSeek founder Liang Wenfeng's comments pointing to an order of magnitude difference between the US and China access to compute - and pre-train scale - were notable. The fact the top Chinese AI labs (including DeepSeek) have nonetheless found ways to stay near the frontier also feels notable.

For observers of the top Chinese AI labs, the counter to AI commoditisation as a negative comment on the AI labs as a collective is that... lower cost players win in a commoditised market. As fast followers, the Chinese AI labs are often able to shortcut exploratory R&D the frontier labs must undertake (see Exhibit 11). The fact AI labs price tokens on a return over useful life (recent DeepSeek comments pointed to a 10-month payback period on hardware spend) means this gives the Chinese AI labs an advantage on an after-training-costs basis. The counter-argument to the "price wars will drive margins to zero" argument boils down to the AI labs' need to fund future training runs. ARR and gross profit growth matter a great deal to investors, but to a frontier AI lab they represent validation from the market - and permission to train the next-generation of AI models.

While periodic bouts of price competition are likely, we'd expect this to be driven by tech advantages (both at the model level and on the inference infrastructure side) rather than the type of zero-sum knife fight we see in mature markets like e-commerce. The size of the future opportunity is so large that training the next-generation model and staying ahead on frontier intelligence almost certainly offers higher ROI than spending to squeeze a peer out of business. Over time, we think most of the top AI labs will rent or build out their own compute, leveraging partners they have more bargaining power over, rather than rely perpetually on the hyperscalers.

The new space race... and the geo-strategic argument for success

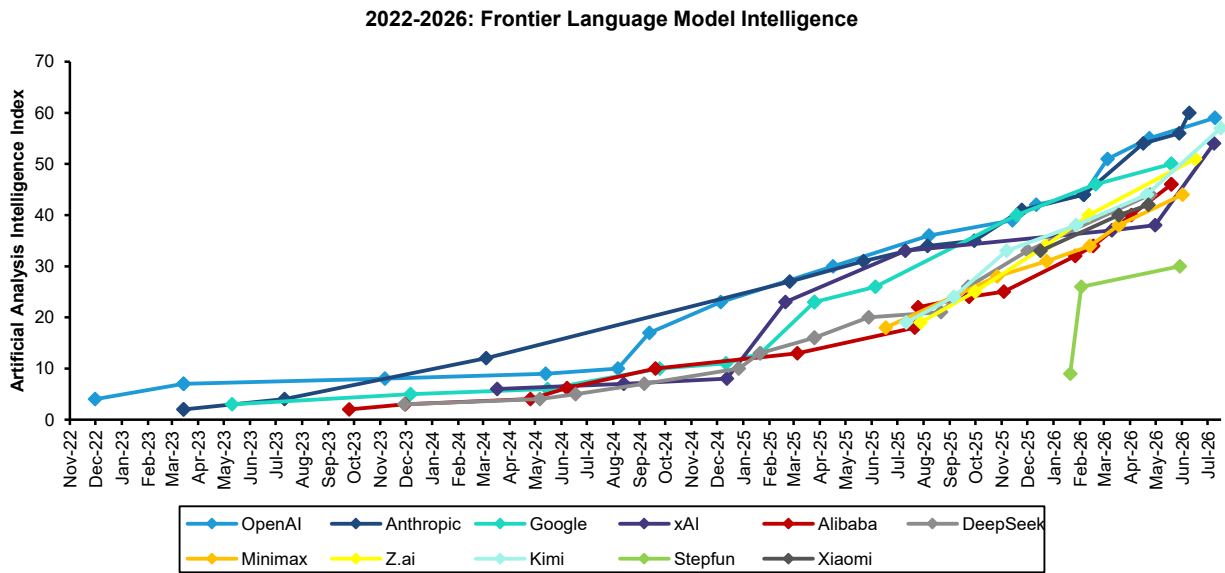
If one accepts that AI represents the new frontier for global geo-strategic competition, with some parallels to the 1960-70s space race, then frontier intelligence represents the obvious end goal for both the US and China across both civilian end uses... and non-civilian end uses. Chinese policymakers have made no secret of their plans to adopt AI as a core pillar of national policy, and support an indigenous build-out of the AI semis and hardware supply chains. The history of strategically important manufacturing sectors in China suggests the build-out won't simply get to demand-supply equilibrium then stop (see Exhibit 12). With a growing range of local Chinese governments declaring support for token factory and neocloud-like businesses, we expect the supply of compute in China to ease over time, in contrast with the 2026 reality of hyperscalers announcing repeated price increases.

Exhibit 13 shows a back-of-envelope framework we refer to on the size of the eventual prize. If cumulative AI capex in China exceeds \$1tn by 2030, the rough 3-to-1 conversion ratio between capex and incremental revenue that we previously estimated from reported Alicloud growth would imply over \$300bn of revenue TAM for AI end-users... or perhaps slightly lower allowing for compute diverted towards physical and other forms of AI. For what it's worth, our bottom-up estimate for AI revenue TAM in China - based on assumptions about per capita AI spend by tier of knowledge workers, adjusted for differences in global wage levels - also lands in the \$200-300bn range in a world where the global AI revenue TAM is \$1-2tn in a few years.

One can debate how much of this domestic revenue TAM becomes captured by consumer AI platforms and applications, or is lost to price competition. Alibaba, Tencent, Bytedance, and Baidu collectively only account for 50-60% of the contemplated total China AI capex in through 2030E, depending on how much public sector build-out comes through. Ergo, even if consumer AI and other parts of the ecosystem occupy a very large share of the total pool (something our Internet platform coverage has never been priced for...), we still think the 2030E revenue TAM for the top Chinese AI labs can be in the \$100-200bn range. Assuming Z.ai stays in the podium positions and takes 20-30% share of the latter revenue TAM should then imply 2030E revenue that's much higher than the \$12bn that street consensus has pencilled in for 2030E (and potentially even the bulls on \$20-25bn). In the overseas market, we've assumed limited long-term access to the US market, to reflect obvious geopolitical realities. But even fairly modest assumptions around adoption in other global regions should imply a meaningful revenue opportunity for the top Chinese AI labs.

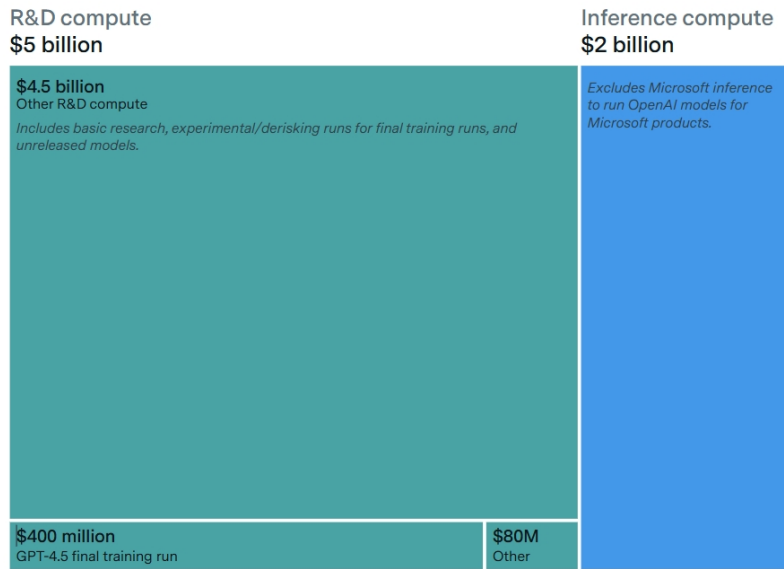
For an alternative perspective on the above TAM estimates, \$150bn/RMB1tn of AI revenue TAM sounds like a big number, but is "only" roughly equivalent to the aggregate annual wages of 3-4mn white collar workers. *That* sounds like a big number compared to 8-10mn software engineers in China, but how many companies in China might *benefit* from replacing their paper pushers, middle managers, and career bureaucrats with AI?

EXHIBIT 10: In the past year the gap between the top US and Chinese AI models has arguably narrowed, with GLM-5.2 and Kimi K3 looking 3-4 months behind frontier



Source: Artificial Analysis and Bernstein analysis.

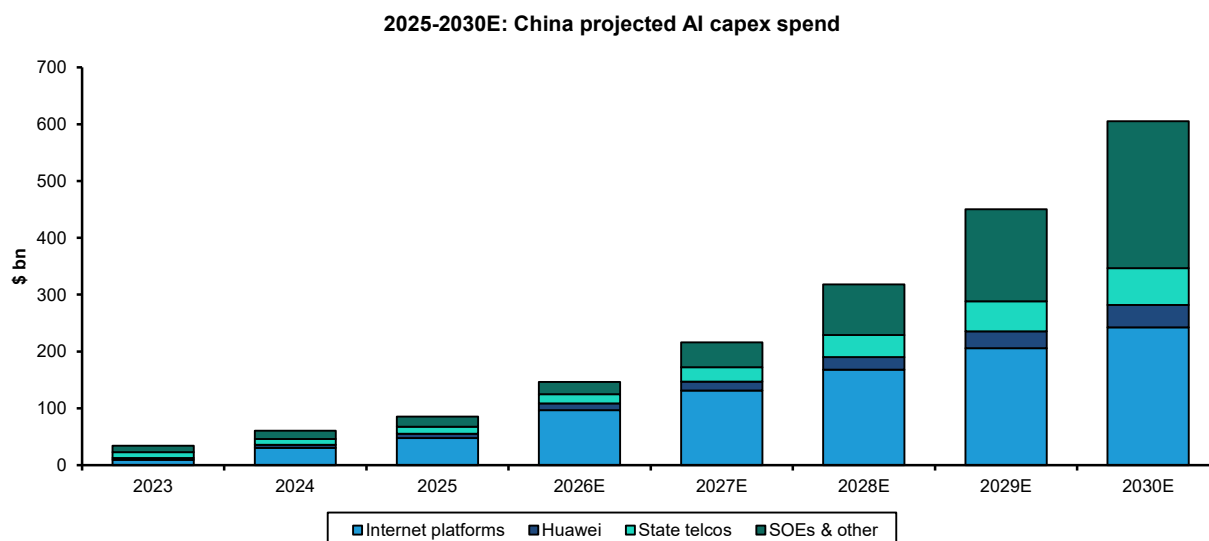
EXHIBIT 11: We expect the lighthouse effect of frontier AI R&D to continue to benefit the Chinese AI labs... most of OpenAI's 2024 R&D was thought to have been driven by exploratory work



All figures are cloud compute expenses.
GPT-4.5 final training run: 90% CI: \$170M, \$890M.
Other models: 90% CI: \$24M, \$435M. Final training runs for other released models from Q2 2024 to Q1 2025, including GPT-4o, GPT-4o mini, Sora Turbo, and post-training for GPT-4o updates and the o-series.

Source: Epoch.ai and Bernstein analysis.

EXHIBIT 12: **There is a clear national interest in the build-out a supply of domestic silicon... our semis colleagues forecast over \$1tn in domestic AI chip sales in 2026-2030E**



China semis covered by Qingyuan Lin.

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 13: **\$1tn of cumulative AI capex should translate to c. \$300bn of domestic AI revenue TAM by 2030, within this we expect the top AI labs to capture meaningful share, and see significantly higher revenue than \$12-15bn consensus**

2025-2029E China AI capex: \$1tn+

Based on discussions with our semis colleagues, we've assumed China AI capex rises from \$150bn in 2026 to \$400bn in 2029E. 2030E capex has the potential to increase this number, though depreciation and compute diverted to physical and other forms of AI likely detract from this number

2030E China AI revenue TAM: \$300bn

Alibaba conversion from capex to incremental revenue has been in the 3-to-1 ratio to date. This ratio may degrade somewhat driven by hardware cost inflation, but we expect China to build out the hyperscaler layer aggressively in the coming years

2030E China AI lab revenue TAM: \$100-200bn

The large Internet platforms we cover account for c. 50-60% of the total contemplated AI capex in China in 2025-2030E. Even if we exclude all of this, assuming a similar capex-to-revenue conversion on the rest should imply a large revenue TAM for enterprise AI

2030E Top China AI labs revenue

The convexity of pricing power for differentiated intelligence means we expect the top AI labs to capture a disproportionate share of the revenue TAM. Under reasonable assumptions we think it's likely the top AI labs generate far more revenue than \$12bn consensus for Z.ai in 2030

Source: Bernstein estimates and analysis.

A MENTAL MODEL FOR AI COMMODITISATION, AND THE OPPORTUNITY FOR "GOOD ENOUGH" AI

We think a generalisable framework for AI model evolution, market competition, and how AI labs as a collective "step off the exponential" on R&D costs hinges on *end user perceptions* of when reasoning capabilities become "good enough", rather than the underlying technology necessarily converging fully in a strict academic sense (see Exhibit 14).

To illustrate, as AI model performance continues to improve, and agentic task completion horizons lengthen, we'd expect a broadening range of tasks to become solvable to an acceptable degree of accuracy, and deployable at scale (see Exhibit 15).

AI-generated images and videos can already be difficult to tell from the real thing. Once agentic purchases of bubble tea, flight tickets, or T-shirts become reliable at scale, the return to doing more R&D on these end uses drops... which should mean AI labs can turn their energies towards more complex tasks.

Do models commoditise... when humans can no longer tell the difference between competing outcomes?

Contrary to what feels like the “typical” theory of model convergence, where AI labs chase a common frontier, and model intelligence eventually asymptotes to a common end-point, our view is that the *gap in perceived intelligence* almost certainly converges once reasoning performance eventually exceed a level where intended users can differentiate between one versus the other (regardless of whether the underlying intelligence truly converges fully).

In recent months, including during the preparation of this report, we’ve consumed several billion tokens per month, including private benchmarking work on various AI models. Looking back, our ability to leverage agentic AI improved meaningfully over time, and our usage of different AI models was educational as far as the *subjective perception* of their reasoning capabilities was concerned. On the other hand, our *subjective perception* of the models’ ability to conduct simple tasks like web searches quickly became undifferentiated... because the task at hand was not complicated enough to lead to wildly differentiated responses.

Extending this analogy, we’d expect most consumer use cases for agentic AI to become “solved” relatively quickly, followed by more deterministic enterprise workloads (e.g. constructing or deconstructing sell-side models) and complex strategic planning. Fields like cybersecurity (e.g. Project Glasswing), advanced engineering (e.g. military hardware supported by autonomous drone swarms), and frontier science (drug discovery, nuclear fusion, space exploration...) will likely continue to demand frontier reasoning capabilities for a long time.

The token rationalisation argument for Chinese models

Our base case is the US AI labs remain the global frontier, and that the convexity of pricing at the frontier means they capture a much larger share of revenues than tasks or tokens (see Exhibit 16 and Exhibit 17). We expect the realities of global geopolitics and issues like data security concerns to mean a meaningful cohort of enterprise customers in Western markets (say some percentage of the Fortune 500, and large enterprises in other developed markets) find the idea of using a Chinese model unpalatable.

Outside this perimeter fence, for a growing envelope of lagging edge workloads outside the US, we’d expect the Chinese AI labs’ lower token costs to prove attractive and help them win share. Compared with Ramp.ai data showing the top 1% of US AI adopters spending \$5k monthly, while the top 10% average just over \$500, we’d guess the next leg of AI adoption will feature employers finding ways to stop the former group from spending \$20k a month... while encouraging the latter to spend \$1-2k (see Exhibit 18). Data from Ramp also showed c. 5% of US enterprise users engaging with Chinese models... led by enterprises whose broader AI engagement has been strongest.

If we simplistically assume the TAM for the Chinese AI labs centres around the domestic China market, plus the rest of the world excluding the US, then that’s still over 70% share of world GDP. If we only include the private sector globally ex. the US (plus all of domestic China) as valid markets for the China AI labs, then “only” about 65% of world GDP is theoretically accessible. Exhibit 19 shows a stricter segmentation of global markets for AI revenue, where we’ve also excluded parts of European and other markets. But even under these assumptions we think 30-35% of the global AI revenue TAM should be accessible to Chinese companies (see Exhibit 20).

Technology wise it felt notable that when OpenAI cut GPT-5.6-Luna prices by 80% it became Pareto frontier for under 24 hours at the lower end of the market... until DeepSeek released the 0731 version of DeepSeek V4-Flash. In contrast, GPT-5.6-Terra prices were only reduced by 20%.

The way off the R&D treadmill

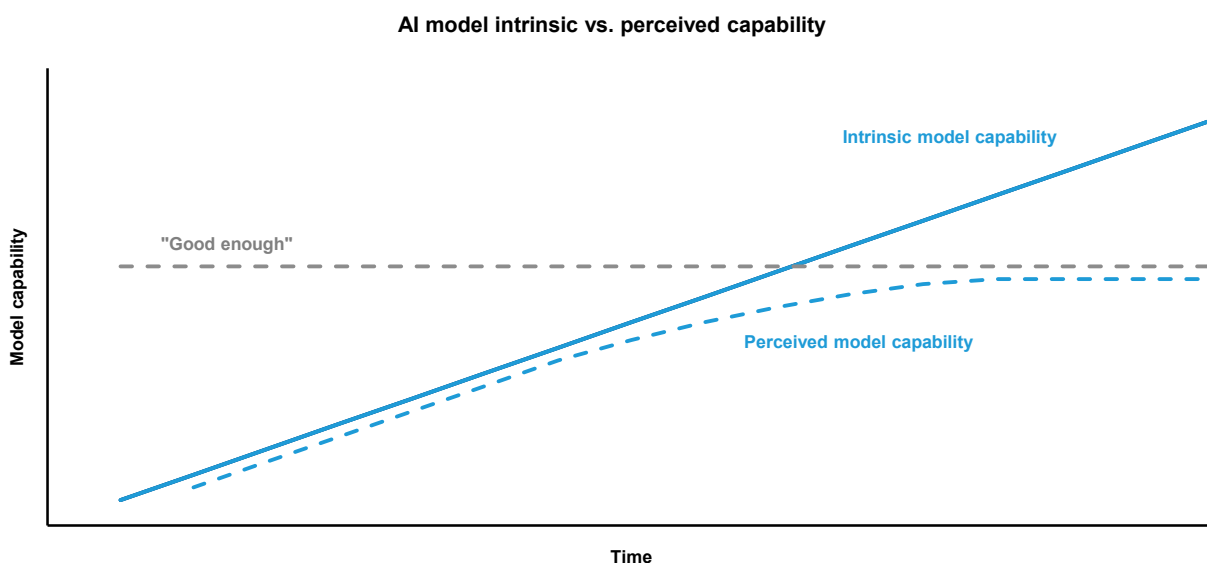
If we’re right about the trajectory of AI commoditisation, the implications for long-term AI lab economics are potentially fairly hopeful. In the foreseeable future, we expect R&D spend among China’s top AI labs to continue to rise rapidly. Over time though, we expect the tapering of annual R&D expense growth to be driven by the ability of these companies to focus investment dollars across a narrower set of model modalities and end uses.

The absolute R&D dollars Z.ai and Minimax will likely continue to trail US peers by a wide margin, reflecting lower developer costs as well as a reduced need to spend on exploratory frontier research. Direct developer costs likely rise much less quickly than R&D spend on development (it’s not clear to us that doubling the headcount of an AI lab proportionately or even positively

impacts R&D outcomes). More importantly, the range of vertical use cases worth spending exponential amounts on for training should narrow over time... meaning model training costs step off the exponential, and AI lab operating leverage improves (see Exhibit 21).

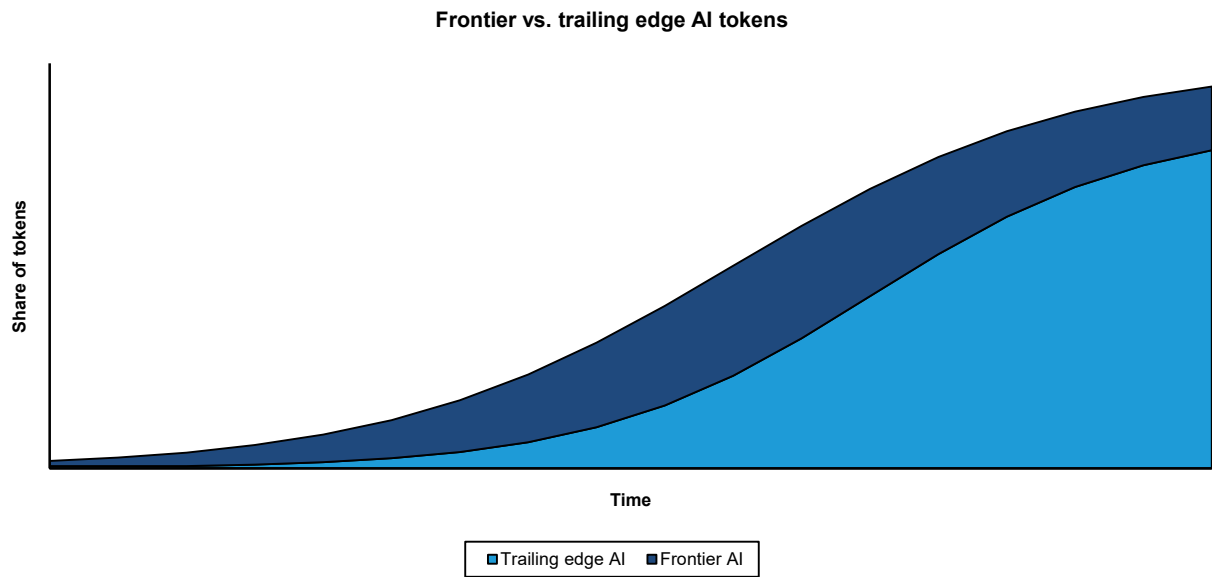
Looking at the rest of the AI lab cost structure, we consider inference margins a product of compute costs and model efficiency, while the rest of the operating expenses stack should remain relatively lean. Marketing spend for example should remain relatively lean in an environment where scarcity value persists for frontier intelligence, and the best models sell themselves to a large extent.

EXHIBIT 14: Our perception of AI model quality increasingly centres around differentiable quality as perceived by humans, as opposed to model capabilities converging in a strict sense



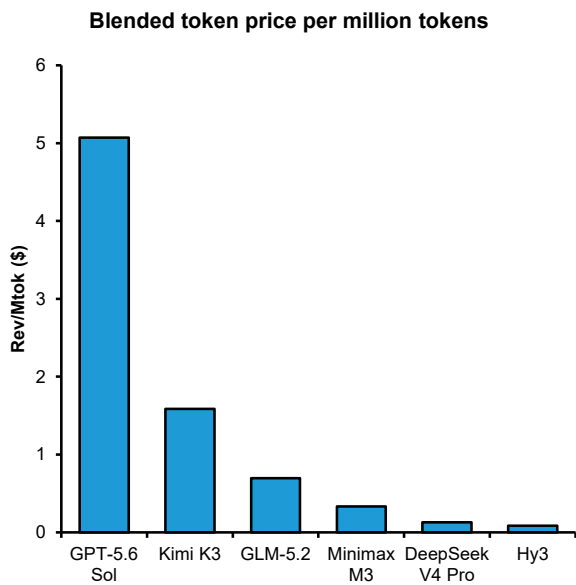
Source: Bernstein analysis.

EXHIBIT 15: **Over time, we'd expect a greater range of agentic tasks to become achievable with “good enough” models**



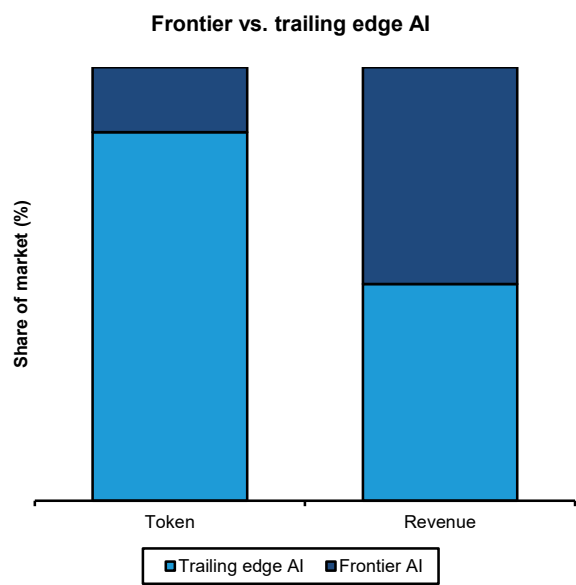
Source: Bernstein estimates and analysis.

EXHIBIT 16: **But the convexity of pricing power at the frontier likely means a greater share of revenue...**



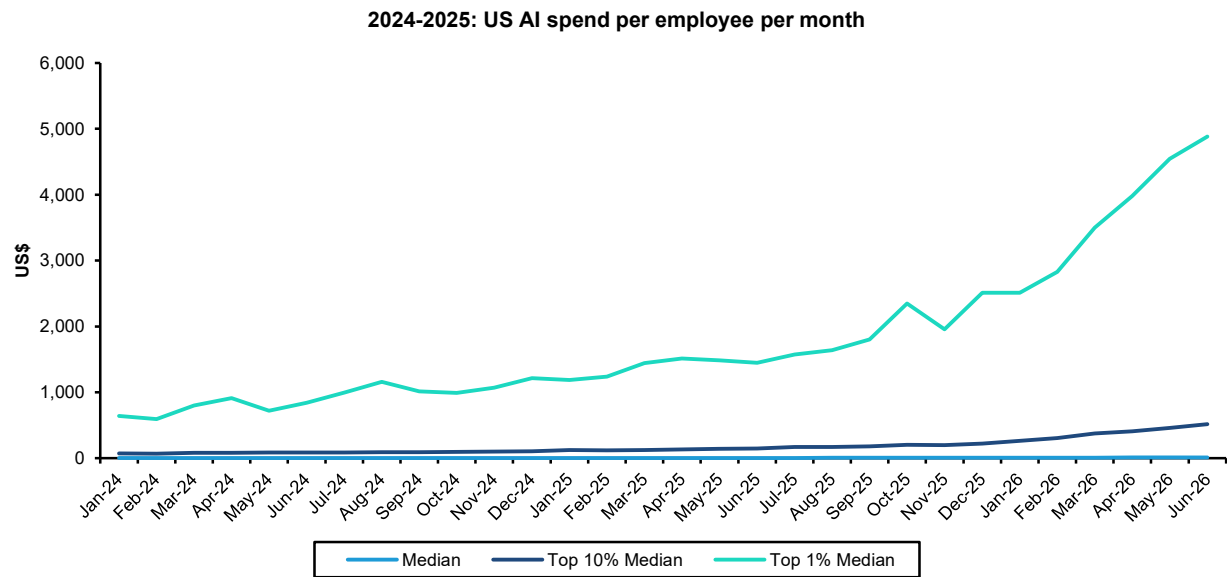
Source: Corporate reports and Bernstein analysis.

EXHIBIT 17: **...accrues to the frontier AI labs... in both the US and China**



Source: Bernstein estimates and analysis.

EXHIBIT 18: Compared with the top 1% of US AI users spending close to \$5k a month, we expect the next leg of AI adoption to need to be driven by broader-based adoption, and a greater focus on token cost efficiency



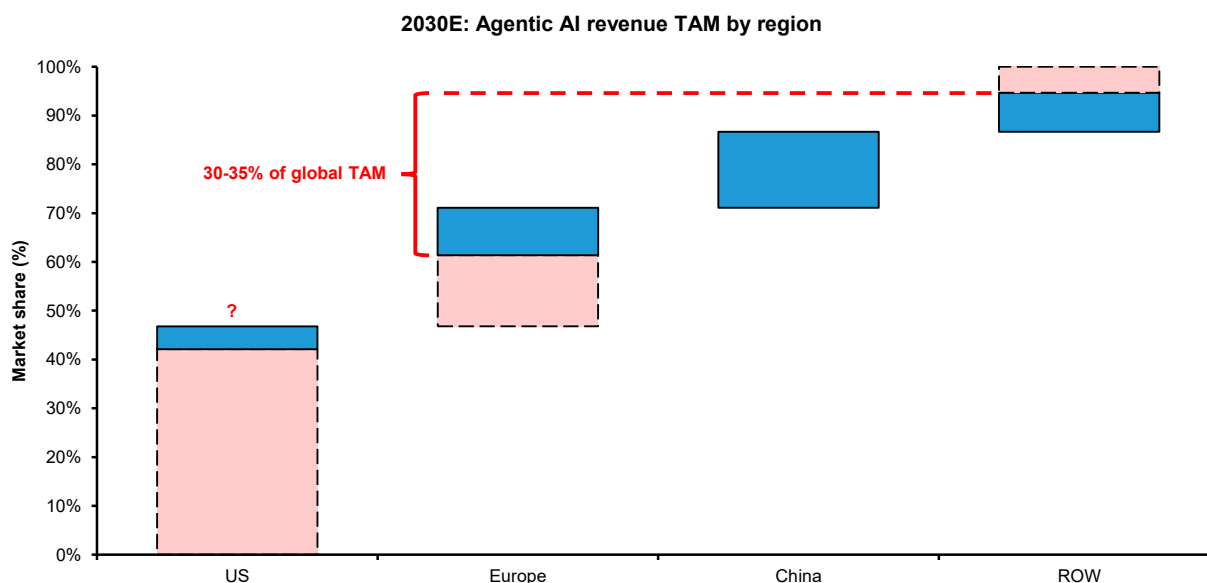
Source: Corporate reports and Bernstein analysis.

EXHIBIT 19: We assume low penetration for Chinese AI models in the US and Europe, though reports like Microsoft potentially engaging with DeepSeek may suggest this is overly cautious

Region	Customer type	Accessibility	Comments
US	Government	Nil	- Assumed inaccessible for geopolitical reasons - Likely reputational and data security concerns associated with use of Chinese AI model - Airbnb on record as having used Qwen model - Microsoft reportedly considering use of DeepSeek to power Copilot - Some access is likely possible, discounted for conservatism - Minimax counts Canva, Twitter/X among its overseas customers - Widespread use of Chinese AI models due to lower token prices
	Large enterprise	Moderate	
	Private sector / start-ups	Moderate	
	Pro-sumers / hobbyists	High	
Europe	Government	Nil	- Assumed inaccessible for geopolitical reasons - Likely reputational and data security concerns associated with use of Chinese AI model, though perhaps not to the same extent as the US - Some access is likely possible - Ongoing use of Chinese AI models due to lower token prices - Likely greater appetite for engagement with Chinese AI labs - Z.ai supporting sovereign AI projects in Malaysia, expanding to Middle East - Assumed lower risk of geopolitical headwinds to adoption, greater sensitivity to lower AI token prices - Ongoing use of Chinese AI models due to lower token prices - Ongoing use of Chinese AI models due to lower token prices
	Large enterprise	Moderate	
	Private sector / start-ups	Moderate	
	Pro-sumers / hobbyists	High	
Rest of world	Government	Moderate	
	Large enterprise	High	
	Private sector / start-ups	High	
	Pro-sumers / hobbyists	High	

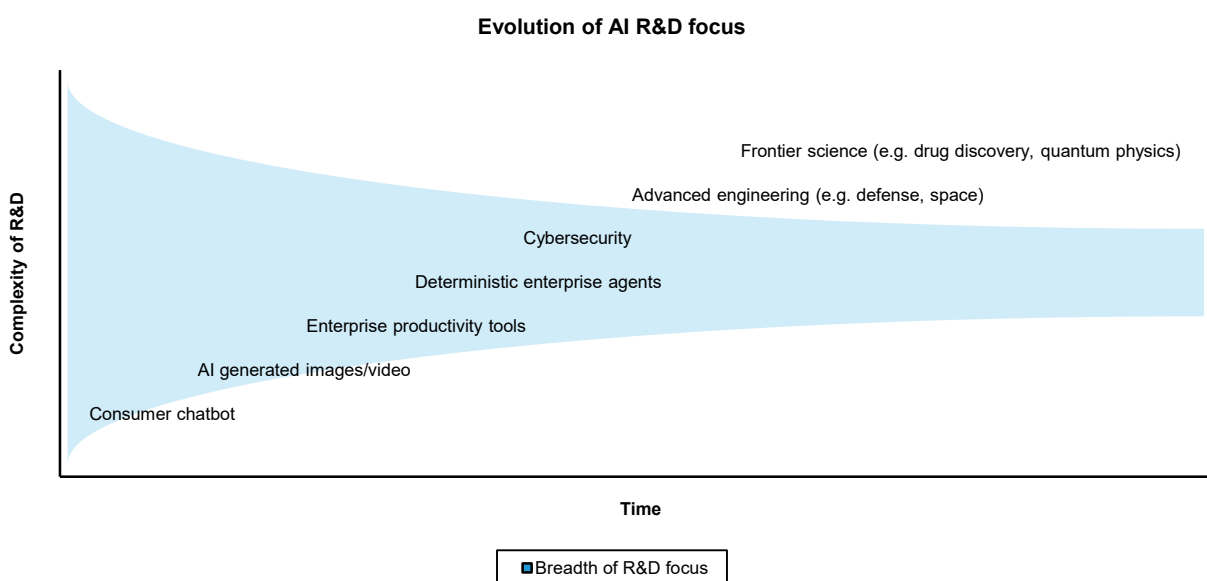
Source: Bernstein analysis.

EXHIBIT 20: We think Chinese enterprise models will be able to access c. 30-35% of the global market for AI revenue, even considering the realities of global geopolitics constraining access to Western markets



Source: Corporate reports and Bernstein analysis.

EXHIBIT 21: Over time we expect a wider range of AI end uses to become “good enough” and therefore no longer worthy of incremental R&D, allowing the AI labs to move on to more frontier end uses



Source: Bernstein analysis.

MODEL QUALITY WILL CONTINUE TO DOMINATE THE DISCUSSION

Model benchmark scores are supplied with every new AI model release, invariably painting the debutante in a positive light. For some investors, many of whom are non-specialists in nature and do not run their own agentic benchmarks, these then become dominant inputs into perceptions of model quality, and AI lab competence. While these documents obviously offer value, we’d suggest investors treat them as marketing materials from the author labs, not least because self-reported benchmarks, differences in infrastructure, and harness conditions can very significantly influence end outcomes. On the margin,

we've generally preferred benchmarks that are (1) newer and therefore less likely to be contaminated; (2) time-gated to only include tasks from after model knowledge cut-off dates; (3) live - meaning the task set is continuously refreshed, preventing memorisation of fixed answer keys; and (4) oriented towards open-ended, long-horizon problem-solving.

For our analysis of AI models, we've chosen to rely on a subset of benchmarks we think are reliable, like DeepSWE v1.1, and Terminal Bench v2.1, OSWorld-Verified, etc. We've also referenced several Artificial Analysis benchmarks, while human preference leaderboards like Arena.ai and OpenLM.ai offer value too (see Exhibit 22 to Exhibit 25).

Our own proprietary benchmarking... and research plans moving forward

Given the potential for even the larger benchmark organisers to succumb to commercial incentives, and for the sake of proof of work on our part (in part to understand the mechanics of AI benchmarking), we've run our own private benchmark tests on a variety of leading-edge models, leveraging DeepSWE v1 (v1.1 wasn't published when we started). While our infra setup is decidedly hobbyist grade, and introduces noise vis-a-vis the official results, our findings have generally been at least directionally consistent with the official results. On some level, we'd almost argue that our test results may be *more* representative of performance levels experienced by API users in the field - in contrast with lab-run benchmark results generated under pristine infra conditions.

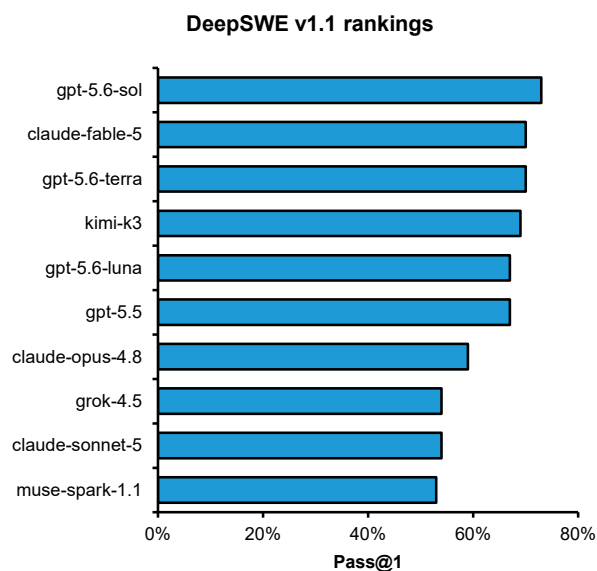
Aside from giving us the ability to claim first-hand validation of model quality, our intention going forward (costs permitting...!) will be to compare future model releases to our existing benchmarking results, to assess model quality after launch - when public benchmark results remain scarce. For what it's worth, the results from our private benchmarking work in the last months were generally confirmatory of GLM-5.2's strong results... as well as the M3 disappointment (see Exhibit 26).

Developer traction as a read on enterprise adoption

One of the difficulties we and investors face when modeling Z.ai and Minimax relates to the tracking of model traction. Channel checks tend to only give a directional steer on how things are progressing, and to the best of our knowledge no reliable third-party data sources exist for ARR. In practice, the information vacuum has left investors dependent on guidance from the companies (for example around compute capacity additions), and given oxygen to what's felt like a near-constant stream of "chatter" on what the latest numbers or go-forward (e.g. 2026-end, 2027-end) ARR expectations might look like. The former is at least official - if infrequent, and the companies presumably guide to beat. The latter adds to the noise surrounding a pair of very volatile stocks.

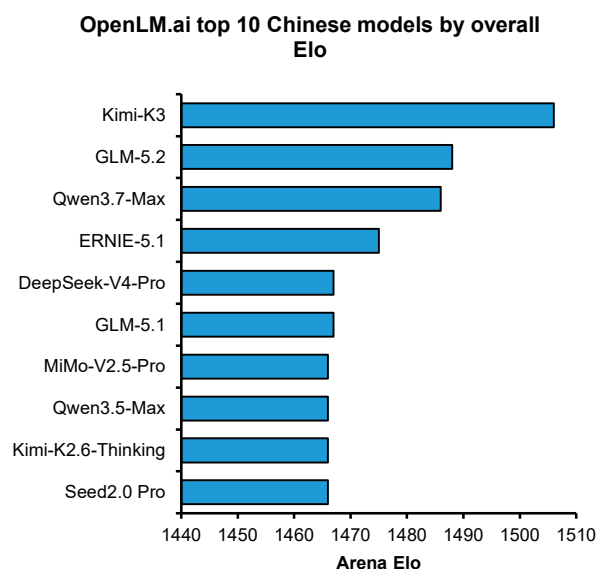
Exhibit 27 shows tracking we've done of China's top AI labs, measured on the basis of domestic and international developer engagement on Alibaba's ModelScope marketplace, and GitHub, which should be more representative of international market trends. This data can be heavily influenced by new model release dates. Nonetheless, the fact Z.ai looks within touching distance of DeepSeek on these metrics has felt notable to us. Kimi's lack of developer traction surprises us, but perhaps reflects its historic focus on knowledge workers and productivity work over heavy-duty coding. Minimax's developer traction has looked more limited.

EXHIBIT 22: **Kimi K3 set a new bar for Chinese models on DeepSWE, but GLM-5.2 looks competitive too**



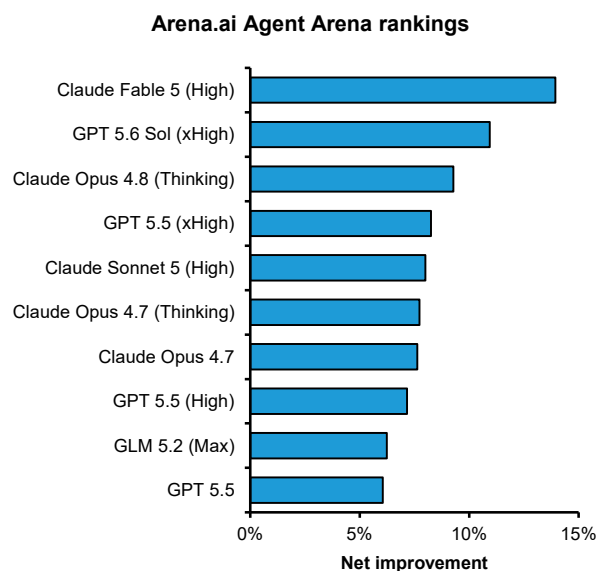
Source: Datacurve.ai and Bernstein analysis.

EXHIBIT 24: **K3 and GLM-5.2 rank top among Chinese model on OpenLM's chatbot arena...**



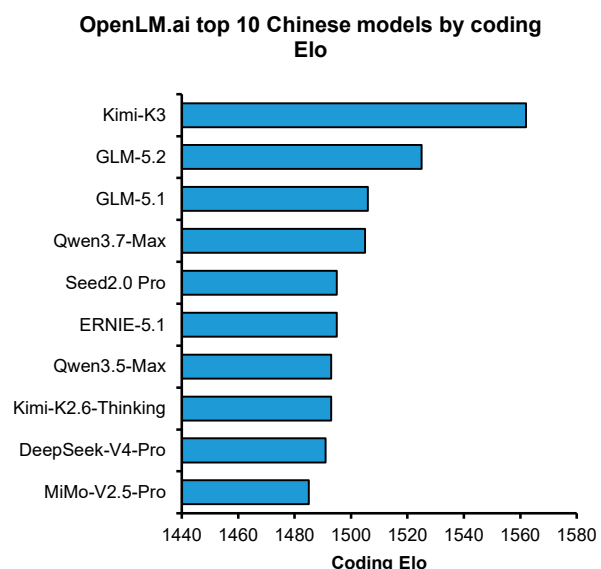
Source: OpenLM.ai and Bernstein analysis.

EXHIBIT 23: **GLM-5.2's Arena.ai Elo looks comparable to GPT-5.5 and Opus 4.7**



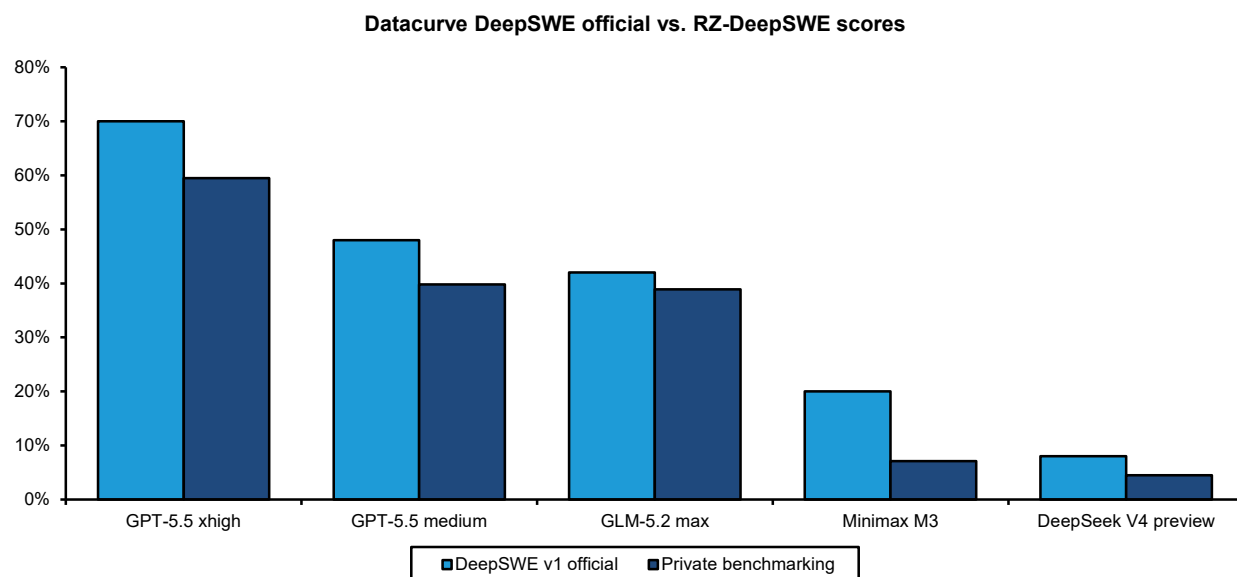
Source: Arena.ai and Bernstein analysis.

EXHIBIT 25: **...measured on overall and coding Elo, and were the highest ranked open weights models**



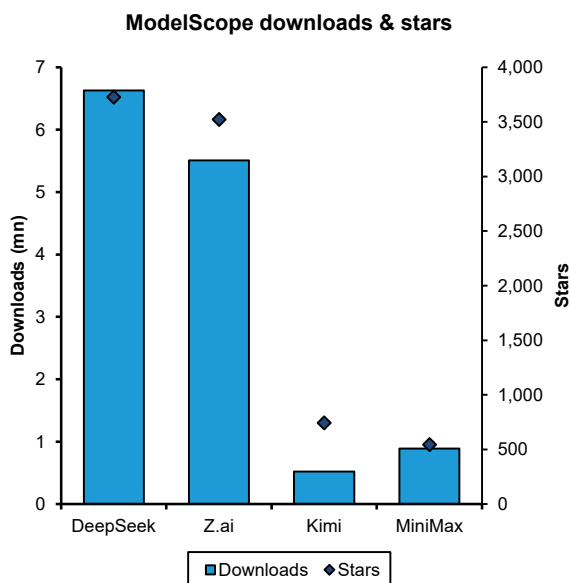
Source: OpenLM.ai and Bernstein analysis.

EXHIBIT 26: On top of day to day use, we've performed private evals of various AI models of interest both as proof of work, and to better understand reasoning performance nuances



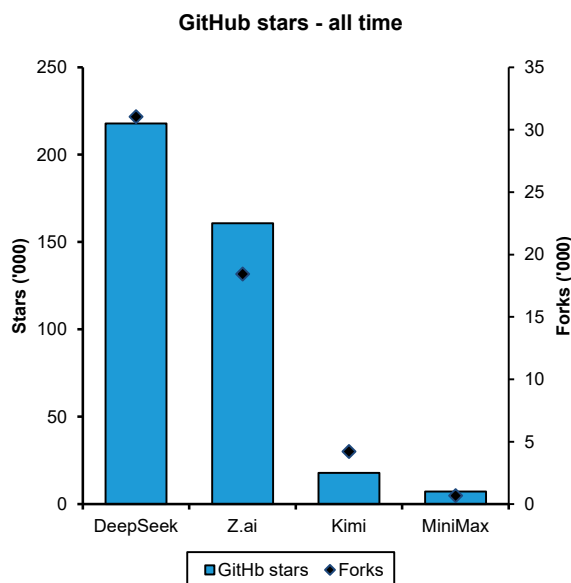
Source: Datacurve.ai and Bernstein analysis.

EXHIBIT 27: To gauge traction for the Chinese AI labs, we've monitored ModelScope downloads in China...



Source: ModelScope and Bernstein analysis.

EXHIBIT 28: ...as well as GitHub stars, as a measure of overseas developer reception



Source: GitHub and Bernstein analysis.

THE FUTILITY OF FINANCIAL ANALYSIS

At the risk of committing sellside research sacrilege, we'd argue that traditional analysis of the near-term financial trends of AI labs like Z.ai and Minimax is most likely not that useful. While the stocks will obviously react to beats and misses, compared with sellside and buy-side boogies, the potential for non-linear change in response to step changes in model intelligence makes linear extrapolation dangerous (this is probably true on the downside as much as on the upside).

Instead of arguing that Z.ai and Minimax trade on massive multiples of near-term ARR or revenues, we think it's more practical to think of the market caps of these companies as an implicit assessment of the likelihood that they'd capture a meaningful share of a very large medium-term AI revenue opportunity. The terminal margins these labs are able to generate will be influenced by whether some of the contenders fall away, leaving the market to settle down into an oligopoly (the more likely scenario in our view, given the research and organisational effort required to stay frontier), or if a persistently large group of Chinese competition drives margins to very low levels. Said differently, it's equally significant that some of the current contenders ultimately fail as the leaders continue to succeed.

Exhibit 29 shows weekly visitation to Z.ai's web pages after the GLM-5.2 release, demonstrating the kind of non-linear growth that can materialise in response to breakthroughs in reasoning capabilities.

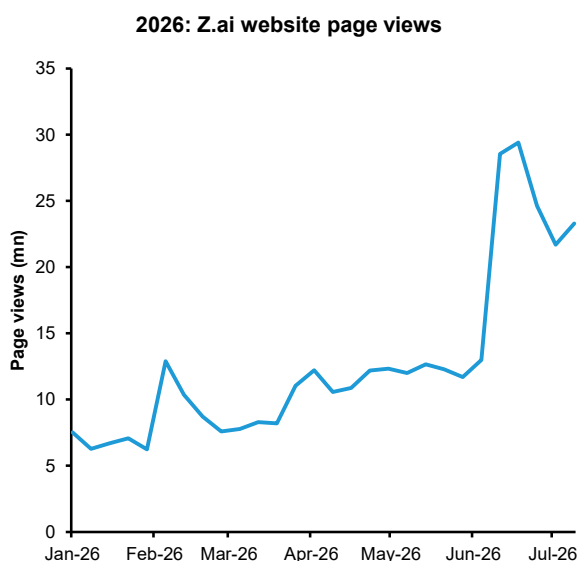
Some thoughts on near-term valuation anchors

For the sake of enabling debate with investors, we've constructed DCFs extending through 2035 - albeit more as a framework to debate the drivers of long-term value creation (e.g. from a "what would it take" perspective) than to pretend we can make accurate projections more than a decade out. The real failure modes in this valuation framework chiefly relate to the labs falling behind on reasoning capabilities, which then causes investors to lose sight of long-term value capture... not dissimilar to what's happened to Minimax in recent months.

More practically, we think comparisons of market caps with private market valuations may offer some value (see Exhibit 30). Z.ai market cap briefly dipped below \$50bn during the post-K3 drop, compared with a similar figure for DeepSeek's last funding round, a \$70bn round that was reportedly then aborted, and Kimi reportedly raising its next round at \$50bn. Another possible anchor for near-term Z.ai valuation that we've heard from investors has included 8-10x 2027-end ARR (which for Z.ai likely argues for a number either side of \$40bn market cap). A turnaround in Minimax's market cap will likely first require the company to deliver a successful M3 Pro model launch.

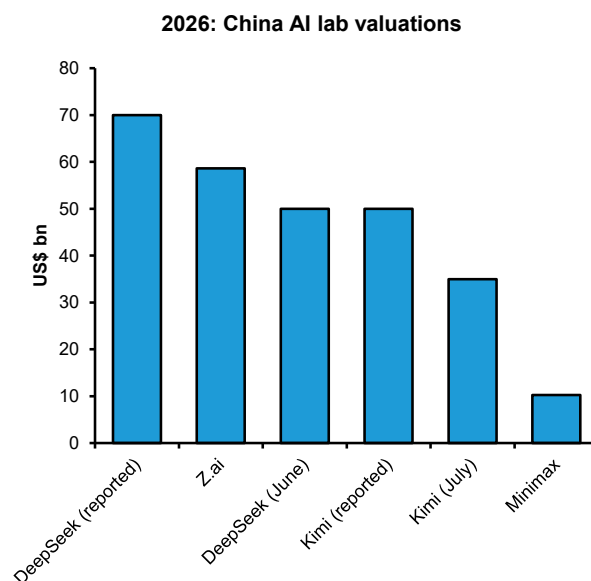
Over shorter horizons, Exhibit 31 describes our perception of the sentiment cycle surrounding AI lab stocks.

EXHIBIT 29: Recent GLM-5.2 interest has been an example of model improvement driving non-linear growth



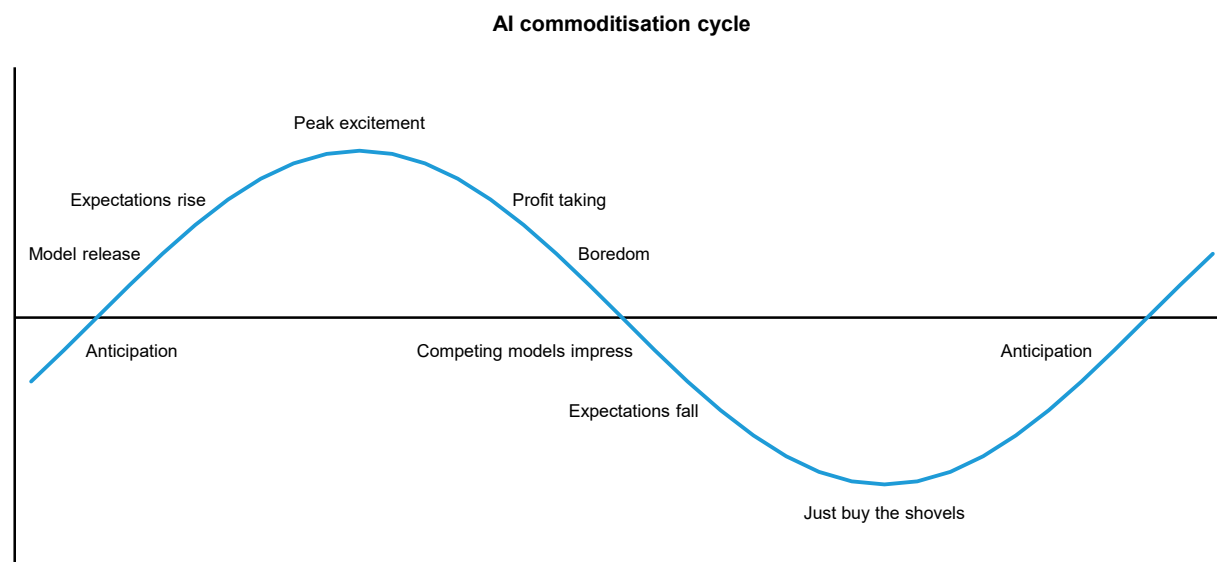
Source: Similarweb and Bernstein analysis.

EXHIBIT 30: ARR multiples aside, comparisons with peer private valuations may offer some value



Source: Corporate reports, Bloomberg, and Bernstein analysis.

EXHIBIT 31: Over more tactical time horizons, we expect the merry-go-round of AI sentiment to persist... while AI fundamentals continue to change rapidly



Source: Bernstein analysis.

Z.AI: INITIATING COVERAGE AT OUTPERFORM (PT HK\$1,350)

We consider Z.ai one of China's top independent AI labs, led by tenured foundational scientists (see Exhibit 32). The company's association with Tsinghua University contributes to pipelines for both talent and domain-specific data. An uncharitable view might argue that Z.ai's GLM-5 family of models has shown more engineering, integration, and RL breakthroughs than during pre-training. But the company's post-training credentials are clearly very strong, and coding has been a dependable performance spike for the company that's driven engagement. GLM-5.2 was the first Chinese model to hit 50 on the Artificial Analysis intelligence index, and the fact Z.ai squeezed Opus 4.7-4.8 level reasoning performance from a 744bn total parameter model drove a viral moment of sorts among online developers and various global AI luminaries. In our discussions, Z.ai pointed to a GLM-5.3 update in the coming weeks, built on the same pre-train as GLM-5.2. The company's next pre-train was pointed towards "September or October", and will be a much larger model aimed at recapturing the China frontier.

On the business end of things, the fact Z.ai serves most of China's largest Internet companies offers logo credibility, as well as a pipeline of data associated with real-world problem-solving. When Tencent banned internal use of Claude earlier this year, Z.ai's models became widely used as the primary alternative. Both our tracking of Alibaba's ModelScope marketplace (as an indicator of domestic developer usage) and OpenRouter token consumption (a proxy for small developer and hobbyist demand overseas) pointed to rapid, non-linear adoption growth following the GLM-5.2 launch.

Level-setting expectations for the next pre-train, and ARR growth

The narrative and valuation reset which occurred in Z.ai's shares after the Kimi K3 release was probably reasonable, following a period where its market cap had briefly exceeded \$125bn. But the view that we'd heard from some investors that Z.ai was "no longer frontier" struck us as premature. Our estimates for GLM-5.2 blended token costs - to simulate workloads heavy on long-horizon agentic tasks - put the model on the Pareto frontier, and both online community feedback and our own day-to-day use of the model have felt like positive references on its ability to remain a "good enough" workhorse model. The fact GLM-5.2 remains on the Pareto frontier after the recent flurry of new model releases also feels notable. We expect GLM-5.3 to deliver another step up in reasoning capabilities... at meaningfully lower effective pricing versus Western frontier models.

That said, the onus is now unquestionably on Z.ai to deliver a competitive next-generation model. In our discussions, the company noted that it had secured enough compute for the next pre-train, and that work was under way. Assuming the "average" 2.0-2.5x jump in parameter count (compared with GLM-5.2 on 744bn total parameters) would imply something like a c. 2tn parameter model, though on the margin, Kimi K3 and Qwen 3.8 weighing in at 2.8tn and 2.4tn total parameters probably

argue for a slightly longer time-to-release, and higher total parameter scale.

Referring to recent media reports, the company confirmed that it was likely to have 1GW of datacentre capacity by the end of this year. Adjusting the \$10bn/GW heuristic we use for OpenAI by relative blended token pricing, we think 1GW of datacentre capacity should coincide with c. \$3bn of ARR run-rate once fully ramped up... but a lower number is likely by December. Notwithstanding the obvious caveats around ARR figures (e.g. last week's revenue times 52...) this likely serves as the "street bogey" for where H2 2026 revenue lands.

Overseas, only some of Z.ai's TaaS partners pay the company a revenue share, while others simply self-host the open weights model. Over time, we'd expect the Kimi K3 template of constraining commercial use to be something Z.ai pursues too - to capture more value from overseas demand.

The high-frequency data we track

The nature of Z.ai's revenues makes high frequency tracking of ARR difficult. Instead, we track a variety of both direct (but piecemeal) and indirect metrics as proxies for Z.ai's progress.

Exhibit 34 and Exhibit 35 show the rise in Z.ai model downloads on Alibaba's ModelScope marketplace, and GitHub stars respectively since June, reflecting the impact of the GLM-5 family of models, including the GLM-5.2 release. Exhibit 33 shows ModelScope downloads across Z.ai's model portfolio since the beginning of June. Exhibit 36 shows the latest SimilarWeb data tracking web page visitation to various Z.ai sites, including a spike in both domestic and overseas web page visits around the GLM-5.2 release.

Exhibit 37 shows our estimates for blended rev/Mtok between GLM4.5 and GLM-5.2, based on headline input, cache read, and output tokens - Z.ai pointed out that the removal of low context pricing (which exists for GLM-5.1 but not GLM-5.2) means GLM-5.2 has 10-20% higher effective token pricing. Exhibit 38 shows aggregate weekly Z.ai token consumption on OpenRouter, which shows the non-linearity of GLM-5.2 usage in recent weeks. Exhibit 39 shows the implied ARR attributed to Z.ai token consumption on the platform, which increases more sharply.

Compared with peers, the fact K3 pricing is more than double GLM-5.2 levels means we think both remain Pareto frontier measured on the basis of Arena.ai coding Elo (see Exhibit 40). Meta's Muse Spark v1.1 is closer to GLM-5.2 in terms of positioning, but remains closed source... and is unlikely to compete directly with Z.ai in the Chinese market. The fact Claude Sonnet 5 posted similar benchmark scores to GLM-5.2 while charging token prices almost twice GLM-5.2 levels felt notable. GPT-5.6-Luna is now cheaper, but unlikely to be as good at long-horizon, enterprise grade tasks.

The fact Hugging Face used GLM-5.2 to defend itself from a security breach caused by OpenAI's autonomous agents (because US frontier models refused to assist) was... interesting.

Growing pains bear watching, but the future remains bright in our view

The one longer-dated reservation we have about Z.ai's business relates to its on-premise business. To quote one of Z.ai's major Internet platform customers, this business stood somewhere between "Palantir like ingestion of company data and production of AI inference" and "onsite Z.ai engineers building functionality on request". We expect these projects to generate a solid rate of repeat business (e.g. GLM-5 customers upgrading to GLM-5.2), even though they're project based and technically distinct in nature. Sovereign projects Z.ai is involved in Malaysia and the Middle East are possible beach heads when expanding further into these markets. On the flip side, Z.ai pointed to higher fulfilment headcount as a contributor to lower on-prem gross margins towards the end of 2025. In 2026 to date, the company has flagged having to compete against hardware cost inflation for finite IT budget allocations. But nonetheless the business was expected to grow in the 50-100% range in the full year of 2026. Over time, we'd expect on-prem to become a smaller part of the overall Z.ai business.

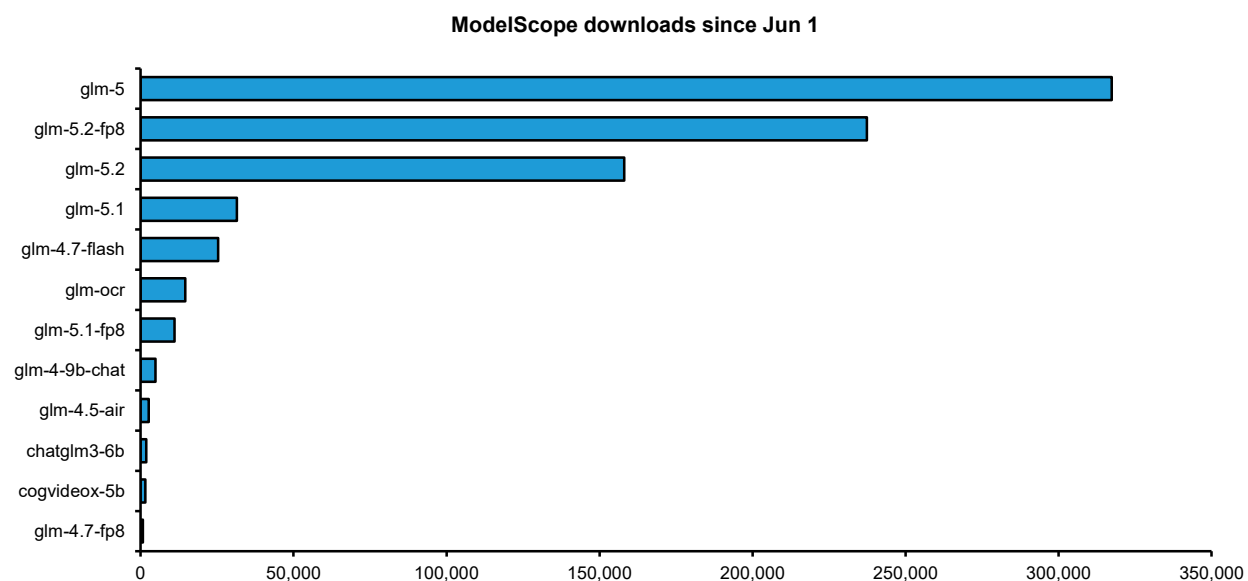
On the API side of the business, the company noted that 50-60% inference margins were achievable when serving API tokens from its own servers. When working with TaaS partners like Infracore though the company incurs some distribution costs that means blended API inference margins are lower. This margin headwind from TaaS partner costs should fade over time as domestic compute availability improves, and the company's revenues increasingly skew towards API revenues... but bears watching for the company's upcoming H1 2026 print.

EXHIBIT 32: We consider Z.ai one of China's top three independent labs, led by a deep bench of foundational scientists

Lab	Researcher	Role	Research focus	Prior Institutions	Citations - all	Citations (2021-)	h-index	i10-index
Z.ai	Jie Tang	Co-founder	Knowledge graphs, social networks, LLMs	Tsinghua; AMiner founder	79,388	61,682	126	461
	Aohan Zeng	Lead researcher	LLM, instruction tuning, agents	Tsinghua	10,227	10,214	19	23
	Xin Lv	Reasoning RL Lead	Reasoning RL, RL Infra, Agentic RL	Tsinghua	7,408	7,228	25	33

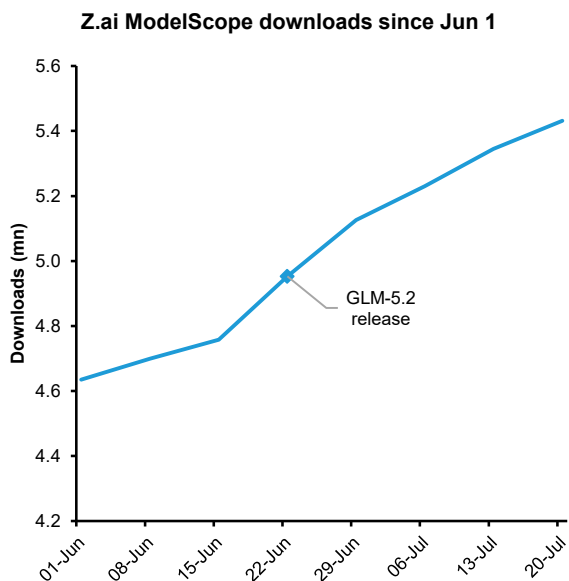
Source: Corporate reports and Bernstein analysis.

EXHIBIT 33: GLM-5.2 now lead cumulative ModelScope downloads since June, while GLM-5 also continued to see solid download volumes



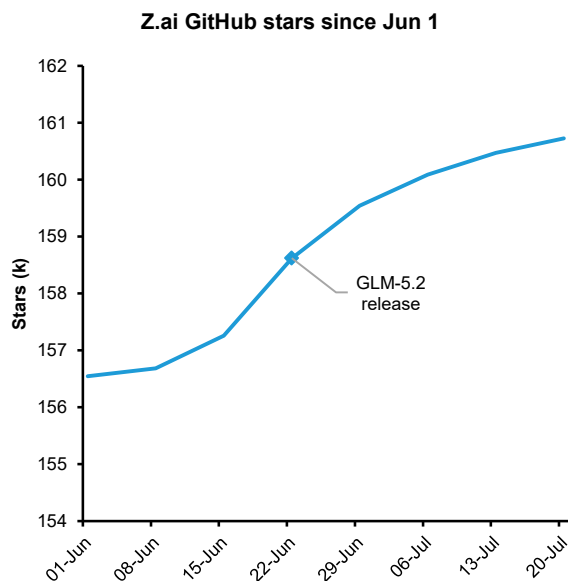
Source: ModelScope and Bernstein analysis.

EXHIBIT 34: Our tracking of Alibaba's ModelScope market place has pointed to steady growth in Z.ai downloads



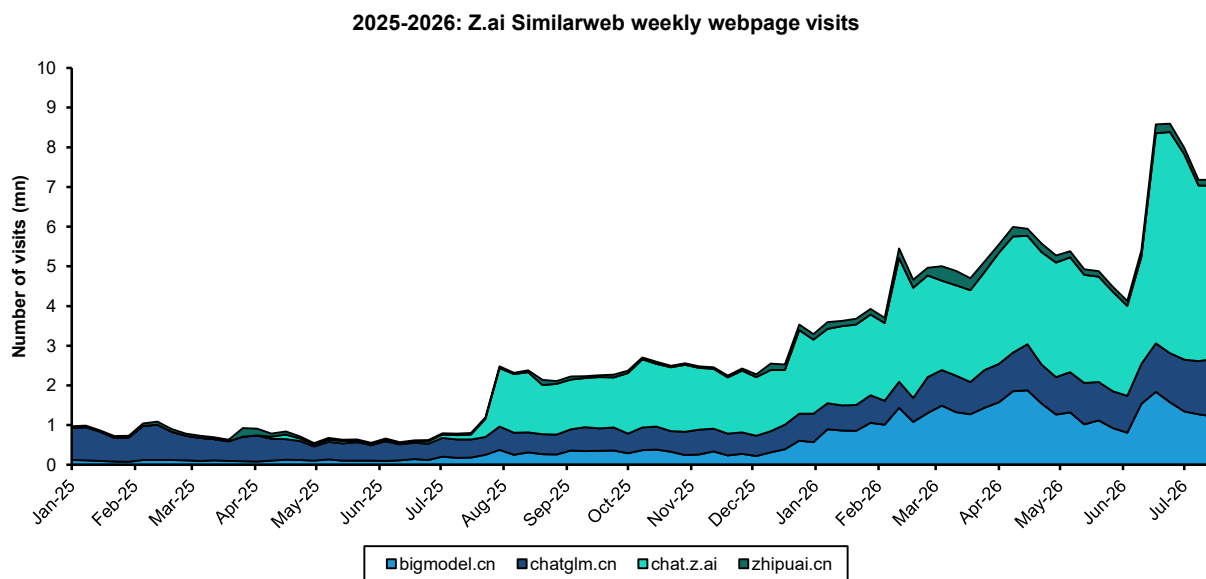
Source: ModelScope and Bernstein analysis.

EXHIBIT 35: The company has also seen steady growth in GitHub stars



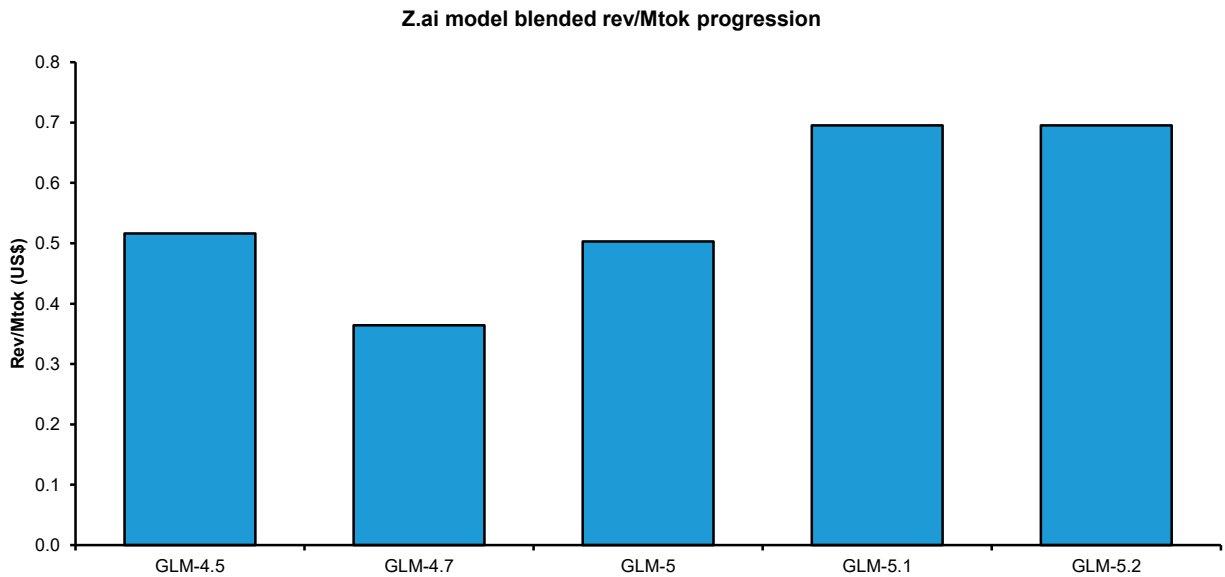
Source: GitHub and Bernstein analysis.

EXHIBIT 36: Z.ai's overseas web page visitation has grown significantly in recent months, especially overseas following the GLM-5.2 release



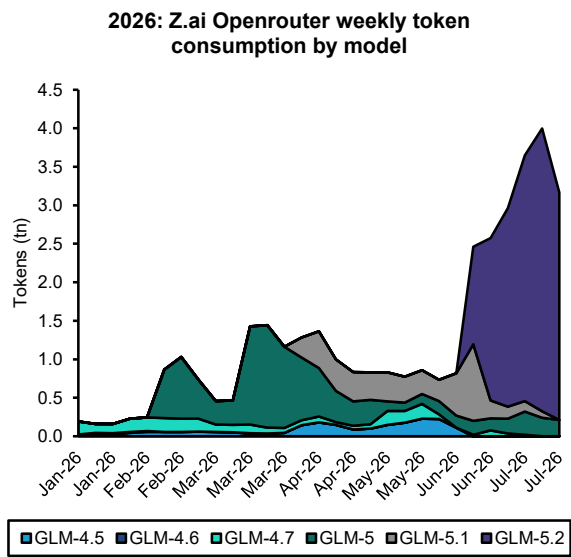
Source: Similarweb and Bernstein analysis.

EXHIBIT 37: **API price increases year to date should have contributed to higher rev/Mtok and inference margins in H1**



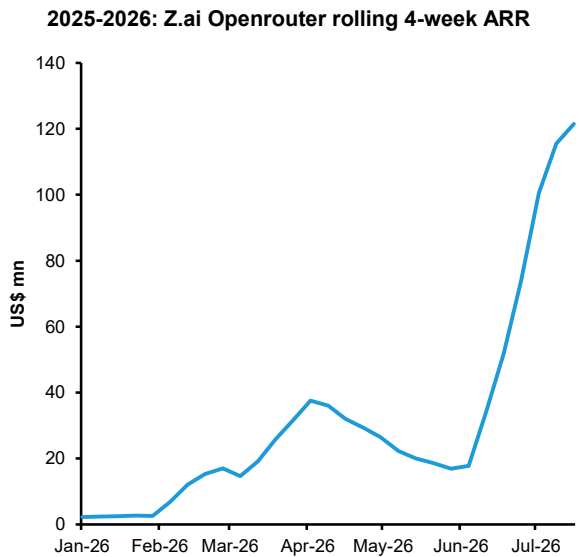
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 38: **OpenRouter data on Z.ai token consumption helps to illustrate...**



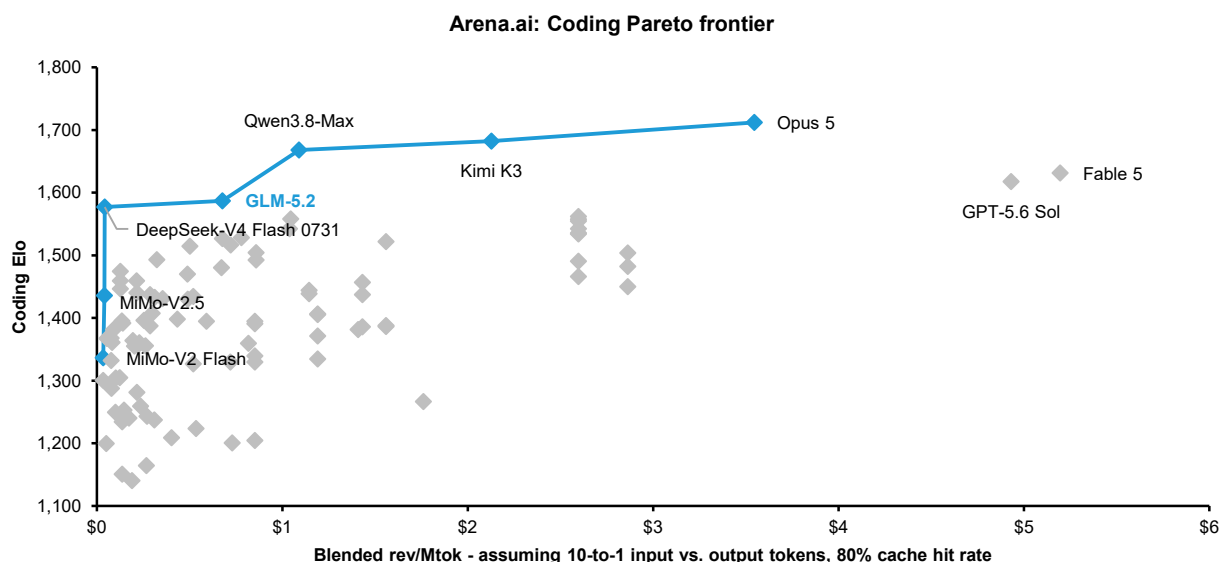
Source: Corporate reports and Bernstein analysis.

EXHIBIT 39: **...the kind of non-linear growth that the GLM-5.2 launch drove**



Source: OpenRouter, Bernstein estimates and analysis.

EXHIBIT 40: Our estimates for blended token pricing puts GLM-5.2 on the Pareto frontier based on coding Elo - while absolute pricing less than half of K3 levels means we think both can co-exist



Source: Arena.ai and Bernstein analysis.

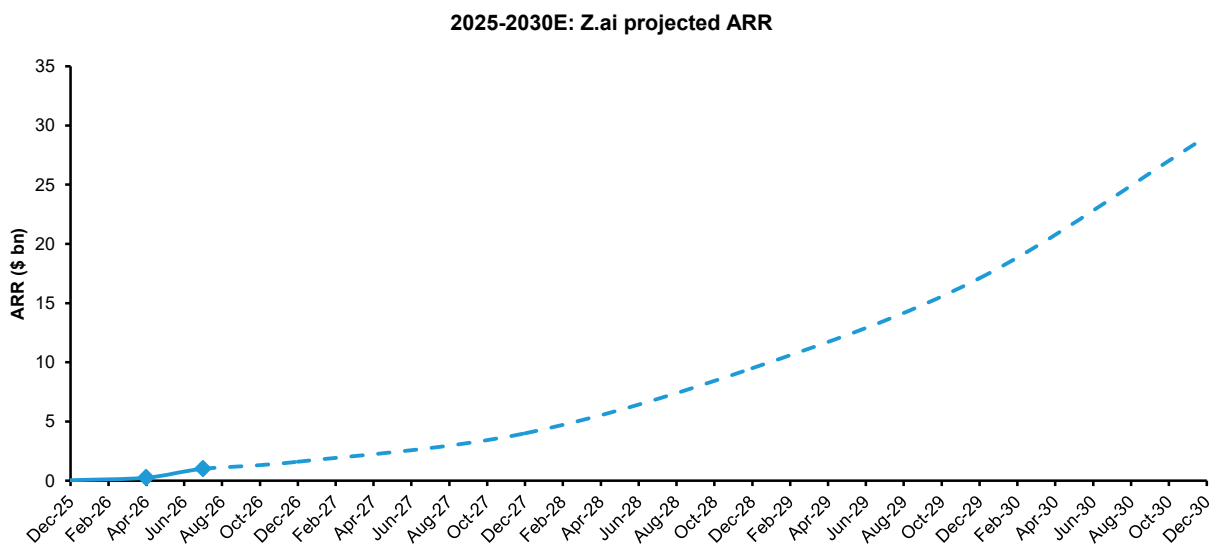
OUR PROJECTIONS FOR Z.AI

For both Z.ai and Minimax, our modelling reflects discussions with the companies over shorter time horizons, but should probably be interpreted directionally in the outer years. Open platform/API ARR for the company hit \$250mn as of late March 2026, and was originally guided to \$1bn by the end of this year (Exhibit 41). Recently news pointing to \$1bn ARR being achieved in July was confirmed to us, while the company's insistence it will remain compute constrained through the year-end (compared against the 1GW datacentre build out) should imply closer to \$2bn ARR by end 2026. We expect greater availability of domestic inference compute to help unlock incremental ARR in 2027 and beyond. Similar to Kimi K3, we expect Z.ai's next model to come with more restrictive commercial licencing terms - aimed at helping the company capture more of the value from international engagement.

For modelling purposes we've assumed \$4bn in API ARR by end-2027. Our revenue estimates in the interim are then estimated by interpolating monthly revenue run rates, and come to RMB5.6bn this year, and RMB18.7bn in 2027. We've assumed close to 70% growth in on-prem revenue in 2026, and close to 50% growth in 2027. We model API gross margins hitting 26.1% this year, then rising to 31.6% in 2027. We've modelled R&D expenses to rise 60% in 2026, per discussions with the company, then 55% in 2027... reaching RMB5.1bn and RMB7.9bn in the two years.

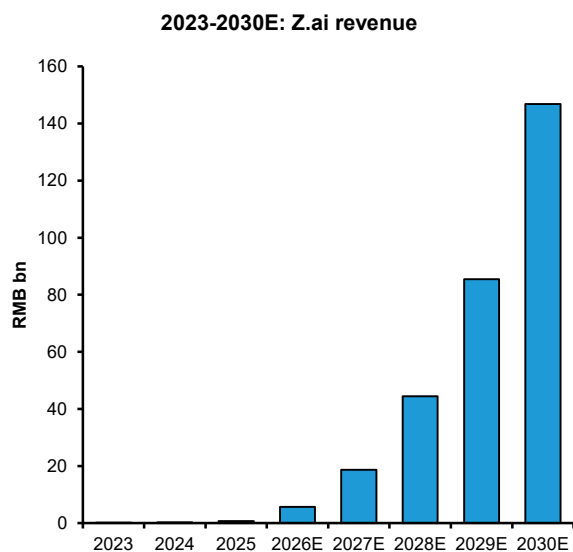
We model Z.ai's operating profit to approach breakeven on a non-GAAP basis in 2028. By 2030, we model the company getting to RMB147bn in revenue, 48.1% gross margin implying RMB70.6bn of gross profit, and RMB31.4bn of GAAP operating profit. Excluding SBC which we've assumed at 6% of revenue, our projection for 2030E non-GAAP operating profit is closer to the RMB40bn range... which drops down to RMB38bn of non-GAAP net income.

EXHIBIT 41: We expect Z.ai's near-term growth to be somewhat constrained by the domestic supply of compute, but expect this to be resolved over time; we model \$4bn of ARR by end-2027



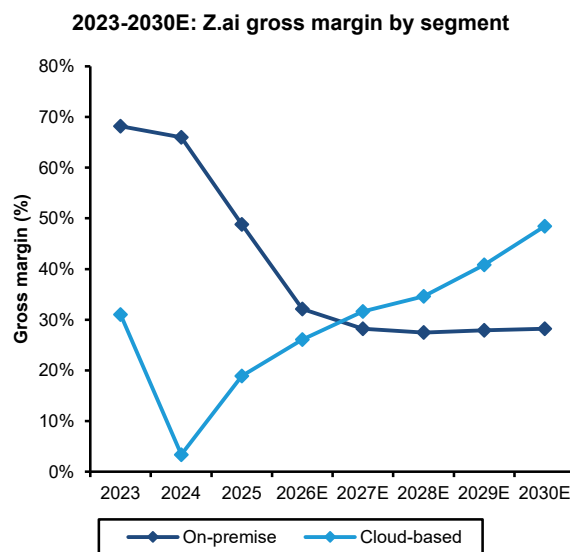
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 42: We model Z.ai reaching RMB147bn of revenue by 2030



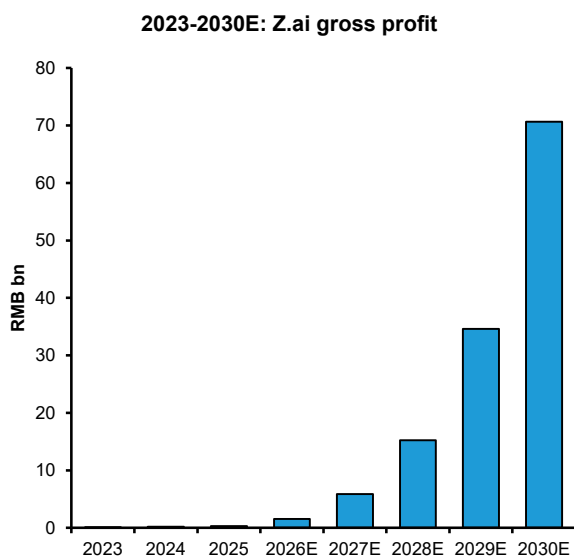
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 43: We've modelled improving API margins, while on-prem margins remain depressed



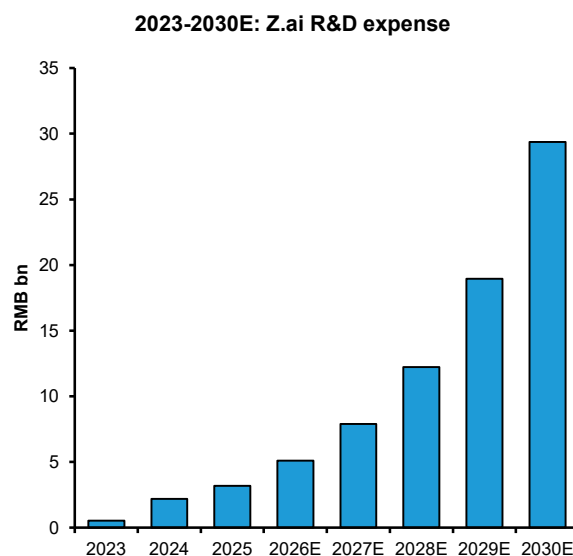
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 44: We model RMB70.6bn of gross profit by 2030, on a 48% gross margin



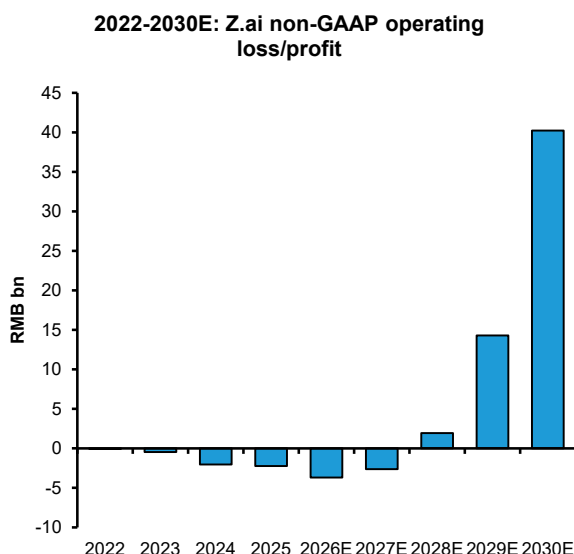
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 45: We've modelled just over 50% CAGR for Z.ai's R&D expenses in 2026-2030E



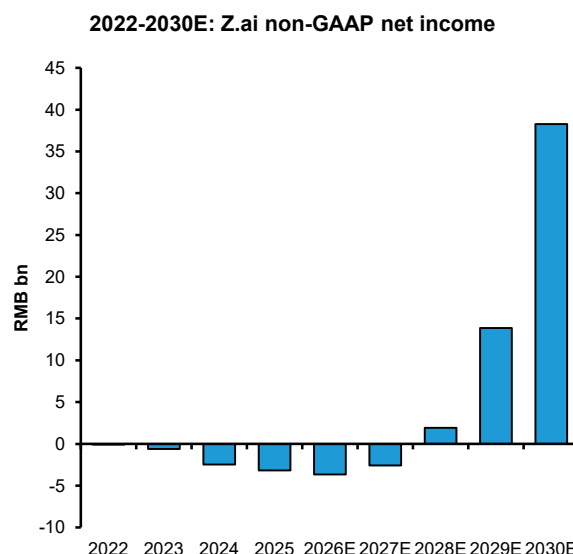
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 46: We model Z.ai to reach breakeven ex. SBC in 2028...



Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 47: Net of SBC we model RMB38bn of Z.ai net income in 2030E



Source: Corporate reports, Bernstein estimates and analysis.

VALUATION

Our HK\$1,350 price target for Z.ai is driven by a combination of approaches, triangulating (1) 8x 2030E PS and 25x 2030E PE - both discounted back at 14% annually; with (2) with our DCF for the company (see Exhibit 48). Exhibit 49 shows our estimates for Z.ai in the next few years, compared with Street consensus. We model Z.ai more or less getting to breakeven in 2028, a year earlier than consensus expectations. We model meaningful new fundraising for Z.ai in 2027, to reflect the company's A-share IPO plans, and equity dilution from SBC in the outer years.

Exhibit 50 shows the pace at which Z.ai's pre-IPO lock-ups expire, allowing its free float to rise. One could certainly argue the increased supply of Z.ai shares in the coming months remains an event to be wary of. On the other hand, if the company can continue to deliver on model intelligence and top-line growth, we'd imagine there will be investors willing to step in.

EXHIBIT 48: **We've built a DCF to illustrate the type of growth required to support our price target for the company**

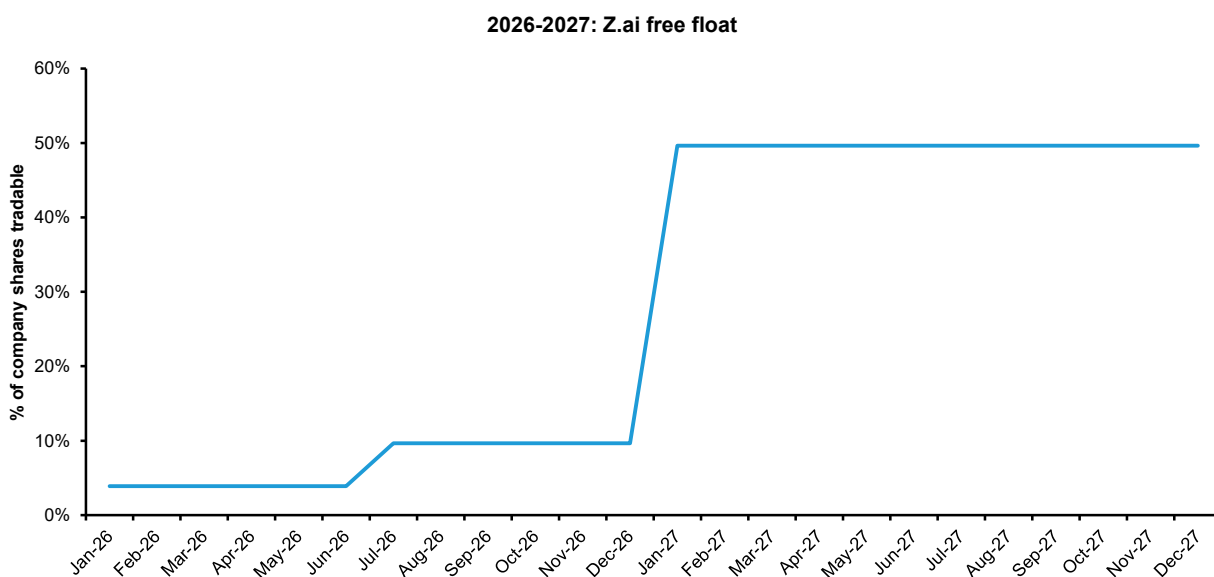
RMB mn	2025	Est. 2026	Est. 2027	Est. 2028	Est. 2029	Est. 2030	Est. 2031	Est. 2032	Est. 2033	Est. 2034	Est. 2035
Revenue	724	5,631	18,663	44,435	85,401	146,813	208,475	260,593	299,682	326,654	336,454
Growth YoY		677.4%	231.5%	138.1%	92.2%	71.9%	42.0%	25.0%	15.0%	9.0%	3.0%
Gross profit	297	1,522	5,855	15,250	34,601	70,622	105,660	138,797	167,347	190,833	205,237
% of revenue	41.0%	27.0%	31.4%	34.3%	40.5%	48.1%	50.7%	53.3%	55.8%	58.4%	61.0%
R&D expenses	-3,180	-5,089	-7,887	-12,226	-18,950	-29,372	-42,472	-56,997	-70,563	-80,018	-82,419
Growth YoY		60.0%	55.0%	55.0%	55.0%	55.0%	44.6%	34.2%	23.8%	13.4%	3.0%
Other operating expenses	-881	-1,184	-1,680	-3,740	-6,479	-9,799	-13,876	-18,309	-22,392	-25,224	-25,981
Growth YoY		34.4%	41.9%	122.6%	73.2%	51.3%	41.6%	32.0%	22.3%	12.7%	3.0%
Profit/(loss) from operations	-3,787	-4,751	-3,712	-716	9,173	31,451	49,313	63,491	74,392	85,591	96,837
% of revenue	-522.8%	-84.4%	-19.9%	-1.6%	10.7%	21.4%	23.7%	24.4%	24.8%	26.2%	28.8%
Less: Income tax	0	0	0	0	-1,376	-4,718	-7,397	-9,524	-11,159	-12,839	-14,526
Effective tax rate %		15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%
NOPAT	-3,787	-4,751	-3,712	-716	7,797	26,733	41,916	53,967	63,233	72,752	82,311
Depreciation & amortization costs	281	283	292	323	398	540	785	1,132	1,566	2,066	2,610
Change in net working capital	704	2,314	4,602	7,304	11,699	16,089	22,847	28,558	32,842	35,798	36,872
Capital expenditure	-11	-56	-187	-444	-854	-1,468	-2,085	-2,606	-2,997	-3,267	-3,365
Additions to intangibles	-12	-59	-140	-257	-398	-570	-714	-821	-895	-943	-960
Free cash flow	-2,824	-2,269	855	6,211	18,641	41,324	62,748	80,230	93,750	106,406	117,468
% of revenue	-389.9%	-40.3%	4.6%	14.0%	21.8%	28.1%	30.1%	30.8%	31.3%	32.6%	34.9%
WACC	13.9%										
Terminal growth rate	3.0%										
Discount factor		0.4	1.4	2.4	3.4	4.4	5.4	6.4	7.4	8.4	9.4
Present value of FCF - per share		-1.9	1.5	9.0	23.3	44.5	58.5	64.7	65.4	64.3	61.4
Sum of discounted cash flows - per share	391										
NPV of terminal value - per share	582										
Enterprise value per share (RMB)	972										
Less: minority interest - per share	0										
Add: adj. net cash/debt - per share	44										
Equity value per share (RMB)	1,017										
Equity value per share (HK\$)	1,180										
Implied 12-month forward price (HK\$)	1,344										
Upside/downside	43.0%										

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 49: **Z.ai: Summary of estimates**

Zipu: summary of estimates	2026E	2027E	2028E
Revenue			
Bernstein	5,631	18,663	44,435
Growth YoY	677.4%	231.5%	138.1%
Consensus	4,159	12,228	27,448
Growth YoY	474.1%	194.0%	124.5%
Bernstein vs. consensus	35.4%	52.6%	61.9%
GAAP operating income			
Bernstein	-4,771	-3,732	-736
Growth YoY	26.0%	-21.8%	-80.3%
Margin	-84.7%	-20.0%	-1.7%
Consensus	-4,985	-5,031	-2,583
Growth YoY	31.7%	0.9%	-48.7%
Margin	-119.9%	-41.1%	-9.4%
Bernstein vs. consensus	-4.3%	-25.8%	-71.5%
Non-GAAP net income			
Bernstein	-3,627	-2,565	1,931
Growth YoY	848.9%	-29.3%	-175.3%
Margin	-64.4%	-13.7%	4.3%
Consensus	-4,553	-4,497	-2,360
Growth YoY	21.6%	-1.2%	-47.5%
Margin	-109.5%	-36.8%	-8.6%
Bernstein vs. consensus	-20.3%	-43.0%	-181.8%

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 50: **Lock-up expiry in January 2027 represents an event to watch for Z.ai shareholders**

Source: Corporate reports and Bernstein analysis.

MINIMAX: INITIATING COVERAGE AT MARKET-PERFORM (PT HK\$275)

Minimax's M3 model was released on May 31, and delivered improvements on M2.7, at least measured on the basis of public benchmarks like Artificial Analysis. That said, weighing in at c. 428bn total and c. 23bn active parameters, M3 remains to date the smallest flagship model across the independent China AI labs, compared with GLM-5.2 on 744bn total and 40bn active parameters, DeepSeek V4 on 1.6tn total and 49bn active parameters, and the latest Kimi and Qwen models on over 2tn total parameters. Performance wise, Tencent's Hy3 model recently showed off stronger reasoning capabilities at lower token prices (see Exhibit 5 1). OpenRouter data pointed to M3 driving a bump in weekly ARR on the platform, but lower token prices meant

the impact on ARR was smaller than the GLM-5.2 bump (see Exhibit 52 and Exhibit 53).

Minimax management now openly acknowledges the M3 disappointment, but for the sake of illustration, Exhibit 54 and Exhibit 55 show an example where we asked Hermes agents running the M3 and GLM-5.2 APIs the same question. Exhibit 56 shows appraisals of both responses from a third GPT-5.5-based agent we used as the neutral validator. Our own day-to-day use of the model pointed to higher hallucination and broader error rates for M3 compared to peers.

Strategically, the company's history of directional pivots comes up frequently in our discussions with investors and industry contacts. At various points in the last few years, Minimax has variously focused on AI companion apps (e.g. Talkie, Xingye), AI video generation (Hailuo), and lightweight agentic coding (M2 and subsequent models), before the most recent pivot to frontier reasoning capabilities. The fact Minimax leadership's contributions to AI literature chiefly came in computer vision has also felt notable to us (see Exhibit 57).

The bull case from here, M3 Pro hopes, and the workload mix drag

The bullish argument from here is that Minimax's shares have round-tripped the post-IPO rally, having fallen some 80% from peak levels in March. If Z.ai trades on the basis of being a potential challenger to the US frontier labs, Minimax's current market cap probably embeds no more than hopes that it *might* recover frontier credibility. The company's upcoming (reportedly 2.7tn parameter) M3 Pro model feels to us like a make-or-break event. The company has been signalling a September to October ETA for launch, similar to Z.ai's next model.

While it's difficult to accurately predict model performance ex ante, it does feel notable to us that a 2.7tn model would be a much larger jump in total parameter scale than has been typical among the Chinese AI labs. Training larger models places significant and non-linear demands on compute and cluster stability, training data availability, and management of issues like expert collapse and load imbalances. Model recipe flaws are amplified at higher total parameter scale, all else equal. In our discussions, Minimax pointed to training progressing smoothly, and loss functions showing encouraging progress, but stopped short of claiming the upcoming model release will be China frontier.

The top Chinese AI labs have on average tended to double their model parameter scale every 6-8 months (see Exhibit 58). Minimax going from M2 (230bn total, 10bn active parameters) to M3 (428bn total, 23bn active parameters) took just under eight months, which felt "normal" compared with peer precedents. Kimi K3 had 2.7 times the total parameter scale of K2 (1.1tn total parameters), and launched around a year after its predecessor. Looking beyond total parameter scale, Meituan's LongCat 2.0 was an example of a trillion parameter model where reasoning performance was similar to the much smaller Tencent Hy3, and feels like something of a cautionary tale on scale for scale's sake.

In contrast with our discussions with Z.ai and Kimi which generally focus more on chasing frontier intelligence, our discussions with Minimax seem to reflect a belief that frontier models were destined to commoditise, and that training a lowest-cost model then optimising infrastructure to better serve intelligence was the more important order of business. This isn't an unreasonable position to have in the grand scheme of things, and Minimax likely has more straightforward access to overseas compute than Z.ai. But then... why build a frontier-scale next-generation model at all?

Competition in the market for low-complexity agentic AI ... and other questions

On some level, we think Minimax's shares represent a call option that M3 Pro might come good. All things considered, several things worry us about Minimax's framing of M3 Pro positioning as a cheap but possibly not-quite-frontier model. The first is training a lukewarm model costs just as much as training a good model. Second, competing on the basis of lower token prices puts Minimax in the most direct competition with DeepSeek, who pioneered several now-widely-adopted architectural innovations to improve KV cache, and whose DeepSeek V4-Flash 0731 one-upped OpenAI's 80% price cut for GPT-5.6-Luna. Xiaomi's MiMo family of models has also shown impressive performance, and adds to competition at the low end of the market (see Exhibit 59). The reasoning capabilities achieved by DeepSeek V4-Flash 0731 (and to a lesser extent Tencent's Hy3) also redefined what's possible in terms of reasoning capabilities despite sub-300bn total parameter scale. Minimax's M3.1 model will be important to watch as a yardstick for RL progress on the 428bn parameter pretrain.

Our understanding from Minimax is that its aggregate ARR figure (\$400mn at the end of May) was split roughly 50-50 between consumer (e.g. companion chatbot apps) and enterprise facing workloads. Web page visitation and mobile app engagement data have pointed to limited Xingye growth, versus better Talkie trends overseas (see Exhibit 60 to Exhibit 62). Within the API business, we highlighted in past research the very high rev/Mtok available to text-to-speech workloads, which we think equates to a price of several dollars per rev/Mtok. The company has declined to discuss the proportion of multimodal workloads within its enterprise-facing ARR, but did acknowledge that multimodal workloads have higher margins than coding. Minimax stands to

face a blended rev/Mtok headwind as agentic workloads grow faster and become a greater share of the company's revenue mix (see Exhibit 63).

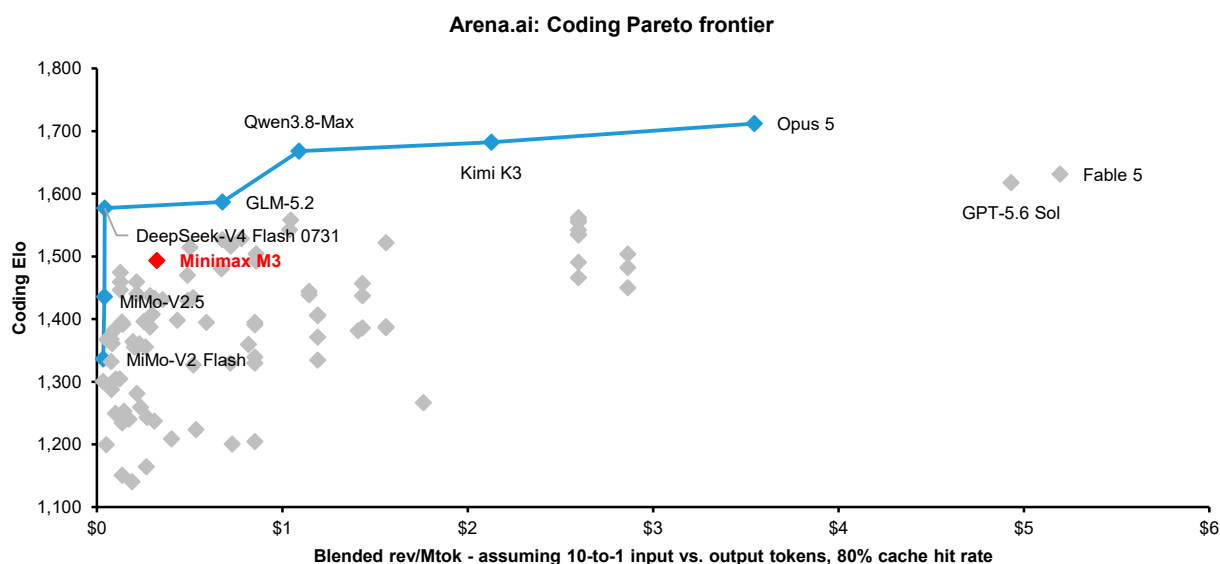
AI video generation plays to Minimax's strengths, but...

Our recent discussions with investors have pointed to signs of renewed optimism, partly driven by the gap between Z.ai and Minimax valuation, but also the company's multimodal growth. The new H3 model showed impressive benchmark performance at a third of Seedance 2.0's prices - Exhibit 64 shows Artificial Analysis' Elo versus price plot for text-to-video generation. We have no doubt AI video generation will continue to drive ARR growth for Minimax. But the realities of the advertising platform market in China, where Tencent, Bytedance, and others are heavily incentivised to prioritise creatives made using native video generation tools means we expect most of the monetisation to have to come from overseas markets. The fact Seedance 2.5 launched almost at the same time as H3 underscores the competition that will likely persist in the AI video generation space, especially when it comes to serious enterprise users.

Compared with agentic coding, where the bull case centres around white collar labour augmentation and eventual displacement, we've long considered multimodal to have a lower ceiling for commoditisation. The fact AI generated video can already be difficult to tell from the real thing raises questions in our view, in terms of the room for product-based differentiation. Once *most* models achieve "good enough" levels of visual fidelity we worry that the main vectors for differentiation become price, subjective yardsticks like "taste"... and price.

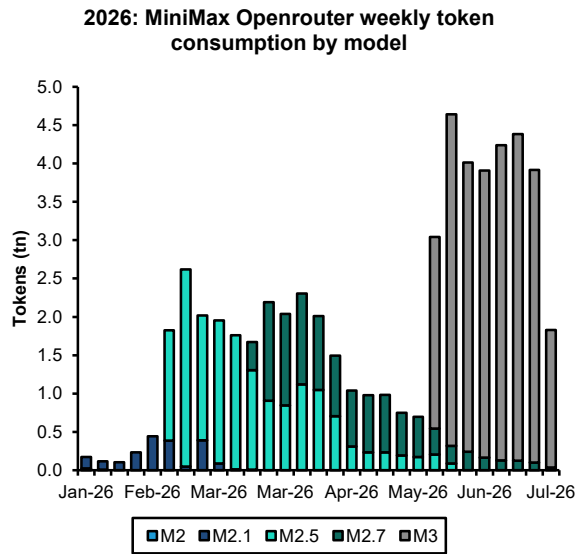
We also think AI video generation has a much lower ceiling for revenue TAM. Compared with white labour productivity work valued in the trillions of dollars globally, the sum of every visual entertainment industry ex. video gaming is likely in the hundreds of billions of dollars globally. Looking out multiple years, Minimax's long-term edge over peers like Google Veo, Bytedance Seedance, or even Kuaishou's Kling is probably a valid question - when the latter have access to large video platforms as data pipeline.

EXHIBIT 51: Our estimates for blended token pricing puts GLM-5.2 on the Pareto frontier based on coding Elo - while absolute pricing less than half of K3 levels means we think both can co-exist



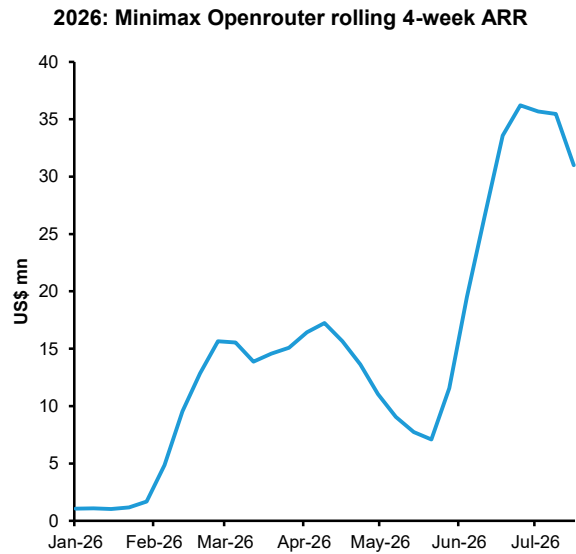
Source: Arena.ai and Bernstein analysis.

EXHIBIT 52: **M3 drove a bump in token consumption on OpenRouter...**



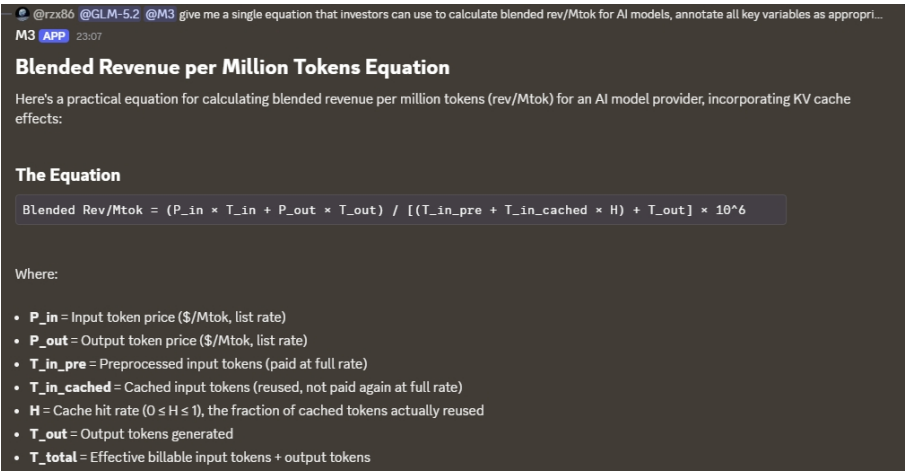
Source: OpenRouter and Bernstein analysis.

EXHIBIT 53: **...though lower token pricing means the ARR impact has likely been lower**



Source: OpenRouter and Bernstein analysis.

EXHIBIT 54: **M3 was a disappointment versus expectations... we asked both GLM-5.2 and M3 to describe the mathematical underpinnings of AI model rev/Mtok...**



Source: Bernstein analysis.

EXHIBIT 55: ... and got a cleaner result from the former, both by our estimation...

@rzz86 @GLM-5.2 @M3 give me a single equation that investors can use to calculate blended rev/Mtok for AI models, annotate all key variables as appropri...

GLM-5.2 APP 23:09

Here's a clean, investor-grade equation:

The Equation

Blended Rev/Mtok = (P_i·N_i + P_c·N_c + P_o·N_o) / (N_i + N_c + N_o)

Variables

Symbol	Meaning	Unit
P _i	Posted price per 1M uncached (fresh) input tokens	\$/Mtok
P _c	Posted price per 1M KV-cache-hit input tokens (prompt-cache reads)	\$/Mtok
P _o	Posted price per 1M output tokens (incl. reasoning/thinking tokens)	\$/Mtok
N _i	Total uncached input tokens billed in the period	tokens
N _c	Total KV-cached input tokens billed (cache reads)	tokens
N _o	Total output tokens billed	tokens

Source: Bernstein analysis.

EXHIBIT 56: ... and according to our GPT-5.5 Hermes agent as the neutral adjudicator

@rzz86 @GPT-5.5 help me grade these response to my question below @GLM-5.2 @M3 give me a single equation that investors can use to calculate ble...

GPT-5.5 APP 23:10

Short verdict

GLM-5.2 is much better.

It gives a cleaner investor-usable equation, separates fresh input / cached input / output tokens correctly, explains the workload mix drivers, and mostly stays inside your constraints.

M3 has serious formula issues.

It mixes token counts, cache hit rate, and pricing in a way that is not internally consistent, and its first equation appears dimensionally wrong.

Grades

Response	Grade	Why
M3	C -	Recognizes input/output pricing and KV cache concept, but the main equation is flawed and confusing. It treats cached tokens inconsistently and seems to confuse cost mechanics with revenue recognition.
GLM-5.2	A -	Correct structure, investor-friendly, includes cache reads explicitly, gives useful ratio form, and explains the business interpretation well. Minor issue: one sentence says "below Pi would be the floor" but the blend is actually above Pi in its example. Also the "single equation" ask was diluted by extra forms/examples.

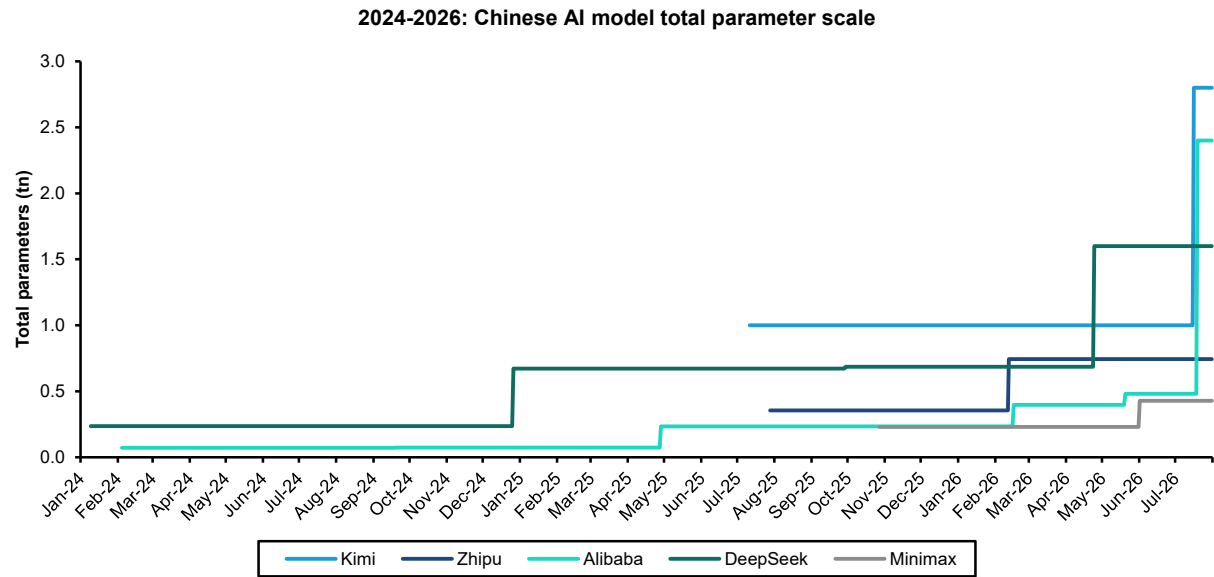
Source: Bernstein analysis.

EXHIBIT 57: Minimax management chiefly come from computer vision and commercial backgrounds, in contrast with Z.ai and other peers

Lab	Researcher	Role	Research focus	Prior Institutions	Citations - all	Citations (2021-)	h-index	i10-index
	Junjie Yan	Founder/CEO	CV, object tracking, deep learning	SenseTime	37,364	27,312	84	127
	Pengyu Zhao	LLM research lead	Streaming media, recommendation algorithms	Hulu	n.a.	n.a.	n.a.	n.a.

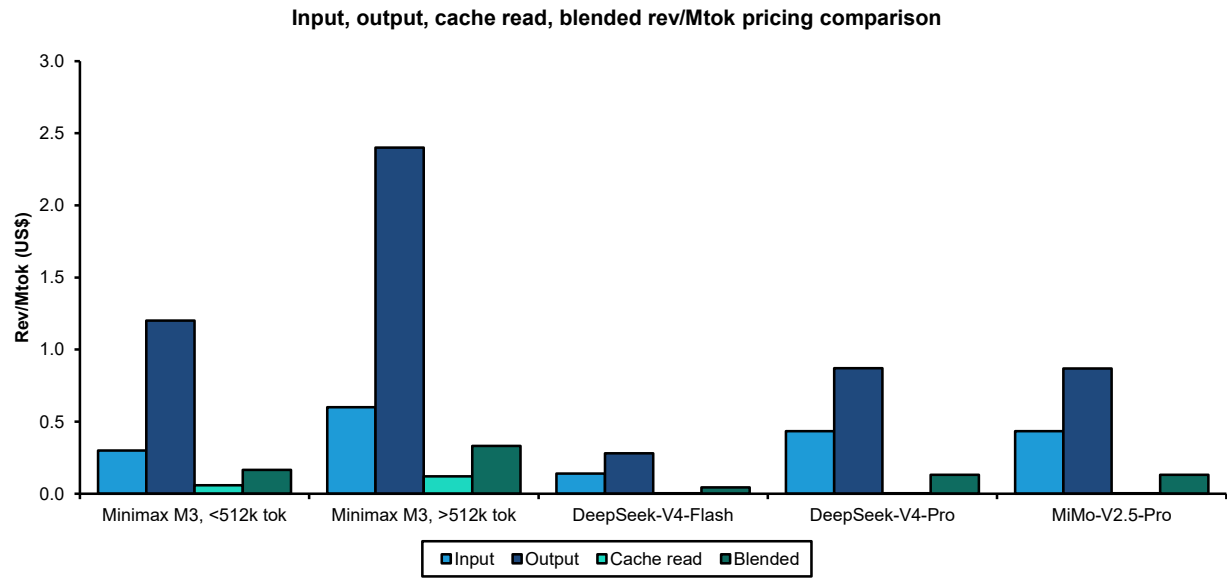
Source: Corporate reports and Bernstein analysis.

EXHIBIT 58: **Reports of an upcoming 2.7tn parameter M3.1 Pro model are potentially encouraging, but would represent a far larger jump than has been typical across the top Chinese AI labs**



Source: Corporate reports and Bernstein analysis.

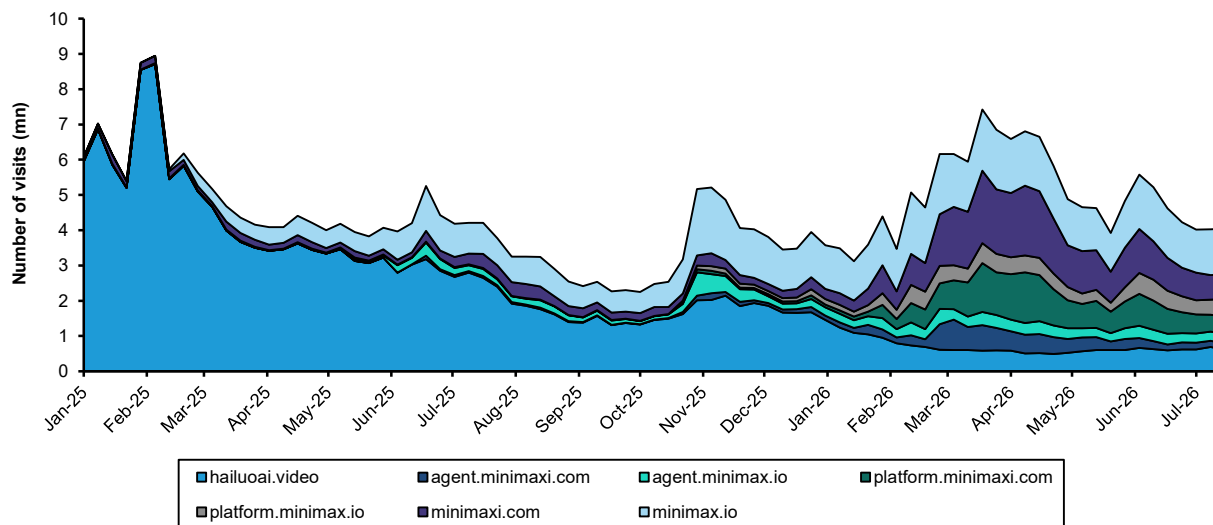
EXHIBIT 59: **DeepSeek-V4 Flash 0731 set a brutal bar to beat at the low end of the market... putting the onus on Minimax to demonstrate it can recover frontier credibility**



Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 60: Web page visitation for Minimax has given investors few reasons for optimism to date... M3 Pro feels like a make or break moment to us

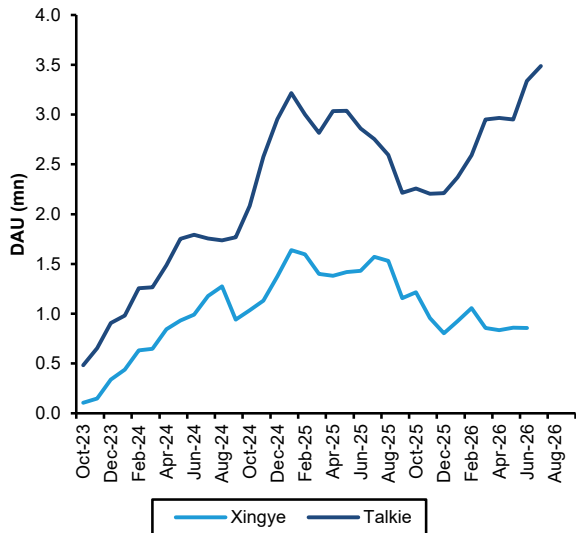
2025-2026: Minimax Similarweb weekly webpage visits



Source: Similarweb and Bernstein analysis.

EXHIBIT 61: High frequency data on Minimax's consumer-focused apps point to limited user growth

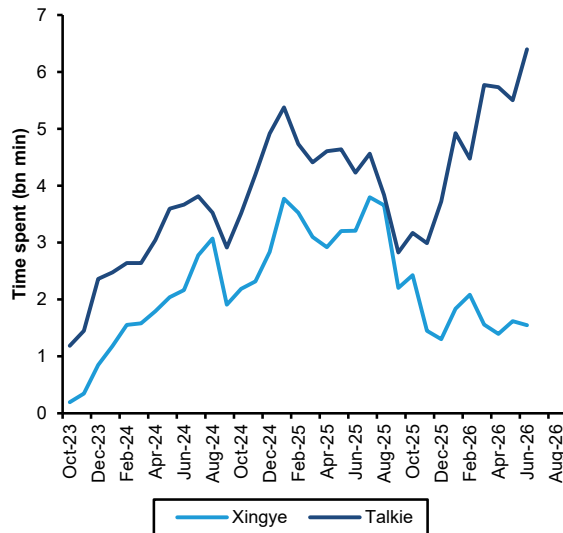
2023-2026: Xingye and Talkie DAUs



Source: QuestMobile, Sensor Tower, and Bernstein analysis.

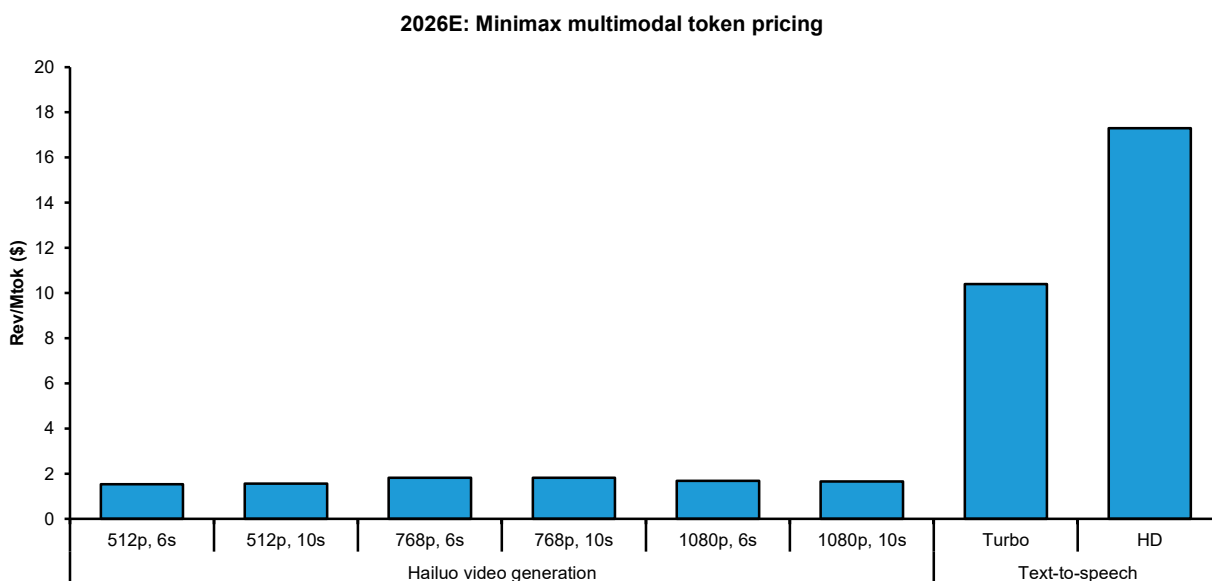
EXHIBIT 62: User growth has looked slightly better, but we doubt this can anchor the long-term bull case

2023-2026: Xingye and Talkie time spent



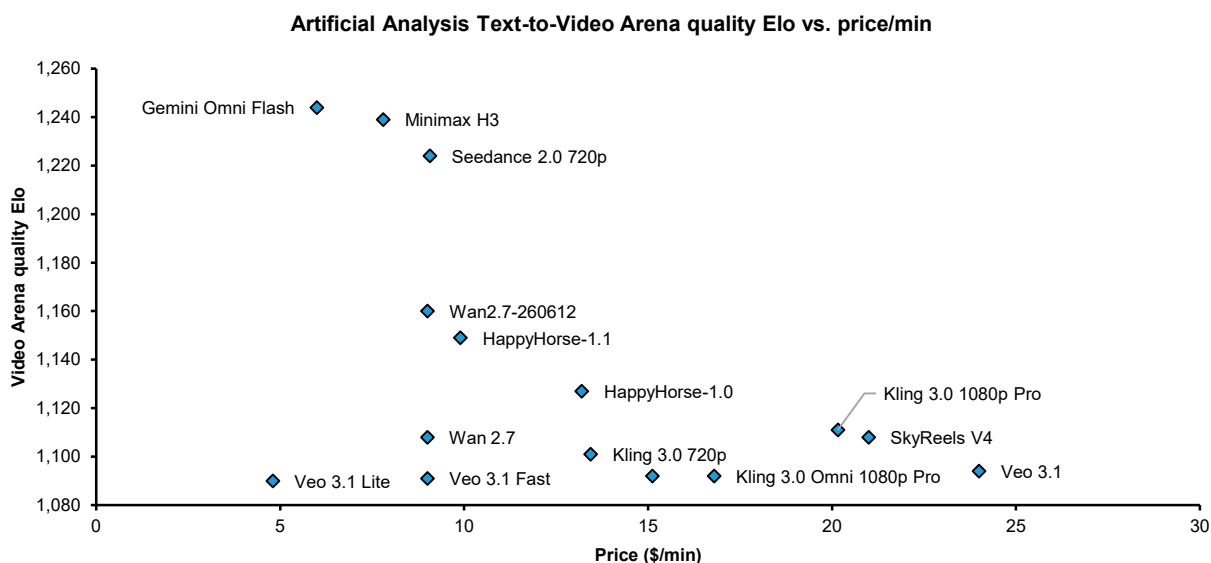
Source: QuestMobile, Sensor Tower, and Bernstein analysis.

EXHIBIT 63: We worry that Minimax will face a mix headwind as its ARR shifts away from multimodal workloads, which generate very high rev/Mtok, and higher than average inference margins



Source: Corporate reports and Bernstein analysis.

EXHIBIT 64: Minimax H3 scored impressively on text-to-video Elo, and likely supports incremental optimism around the stock, but we can't help but feel AI video generation has a relatively low ceiling for commoditisation



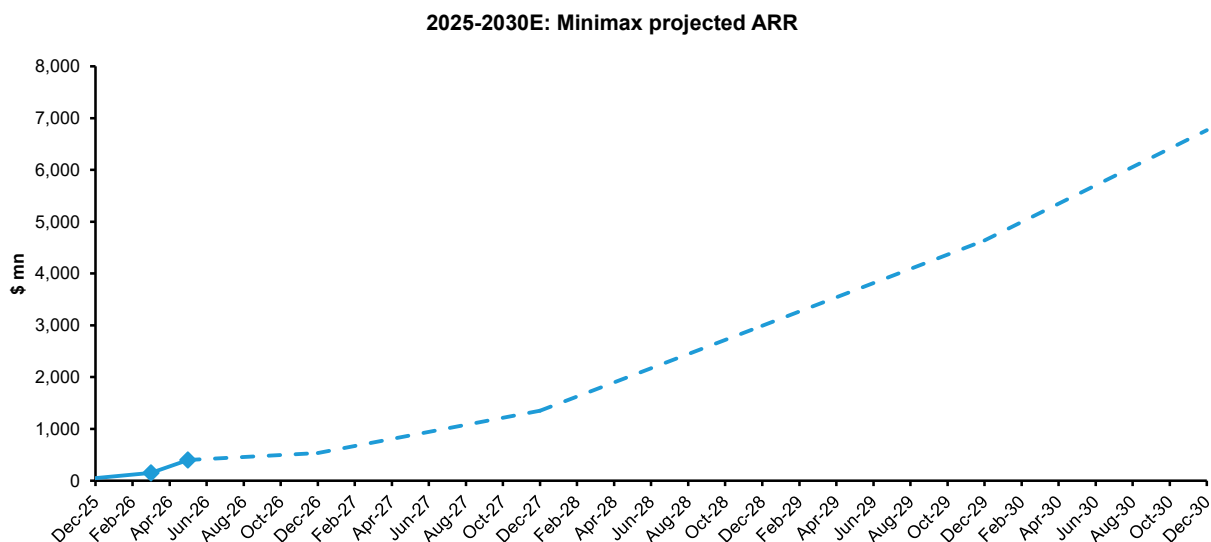
Source: Artificial Analysis and Bernstein analysis.

OUR PROJECTIONS FOR MINIMAX

Our revenue estimates for Minimax reach \$419mn in 2026, to \$1.0bn and \$2.2bn in 2027 and 2028... and \$5.6bn by 2030. For the latter year we model \$2.4bn of gross profit on a 43.4% margin, which drops down to \$159mn of non-GAAP operating profit, reflecting deleverage from high R&D costs. In absolute terms we model \$2.3bn of Minimax R&D spend in 2030E, versus close to \$4bn for Z.ai.

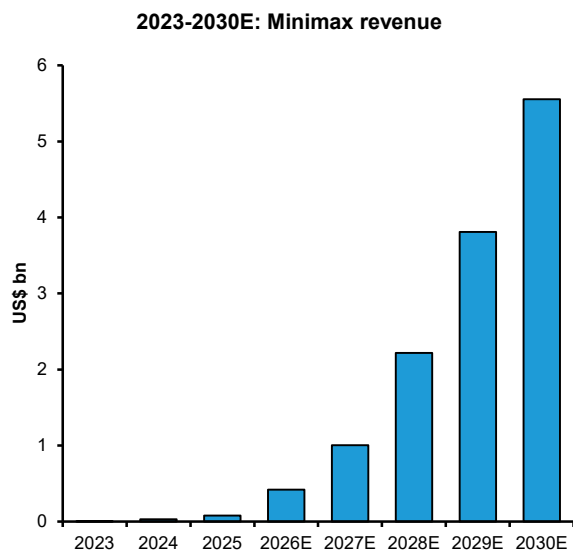
Minimax's decision to stop reporting segment level revenues and margins has been a source of some frustration for us, as we think it obscures several important moving parts that would help to explain the evolution of the business.

EXHIBIT 65: In our modelling for Minimax we've assumed that M3 Pro lands in H2 2026 and is capable, and the company gets to \$1bn of ARR by year end



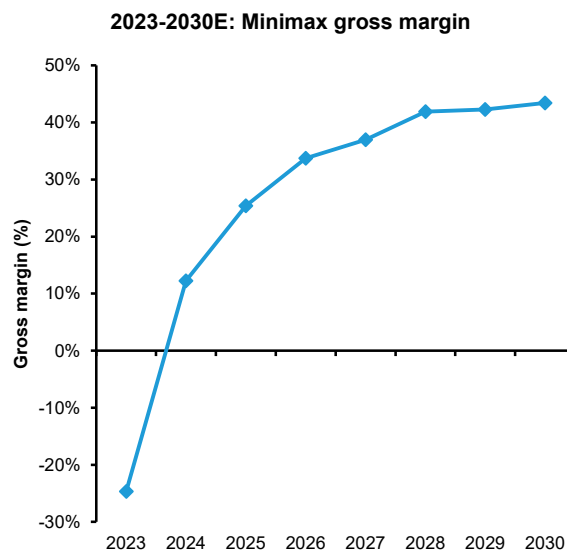
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 66: We model \$5.6bn of Minimax revenue by 2030...

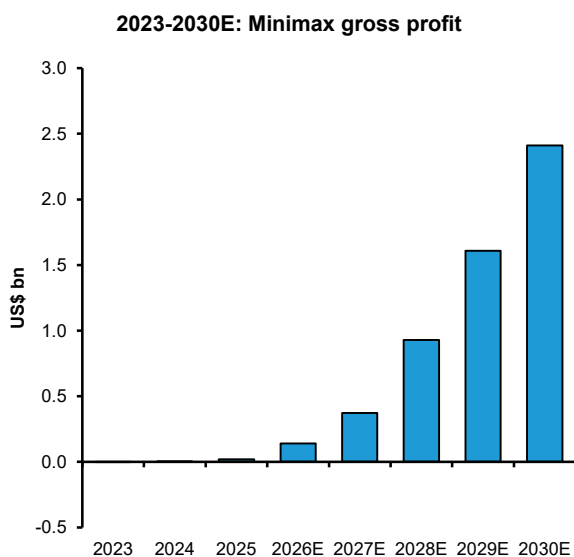


Source: Corporate reports, Bernstein estimates and analysis.

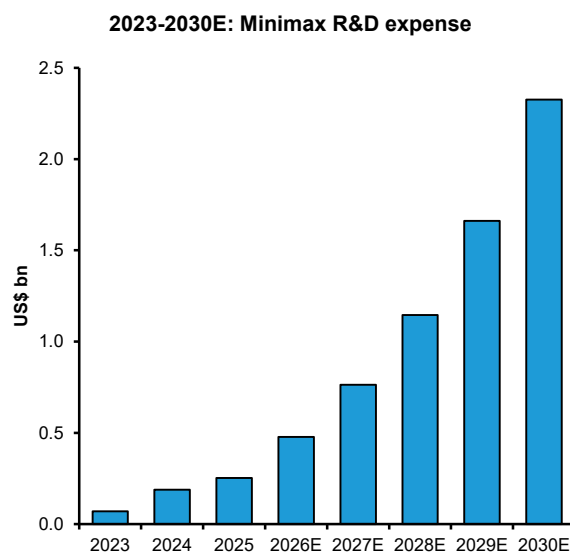
EXHIBIT 67: We model gross margins settling slightly over 40% medium term



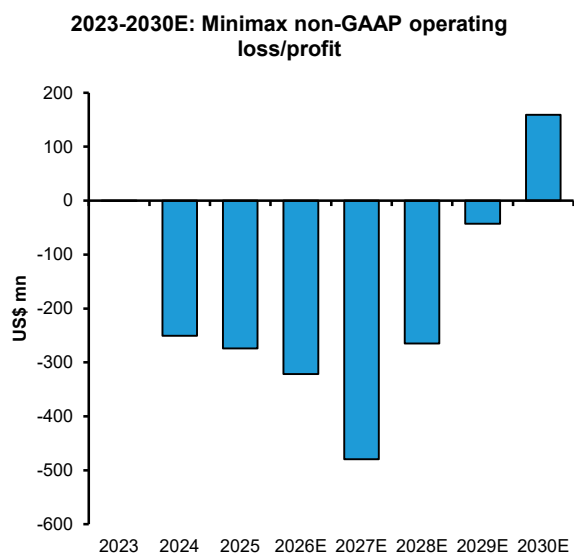
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 68: **We model \$2.4bn of gross profit by 2030...**

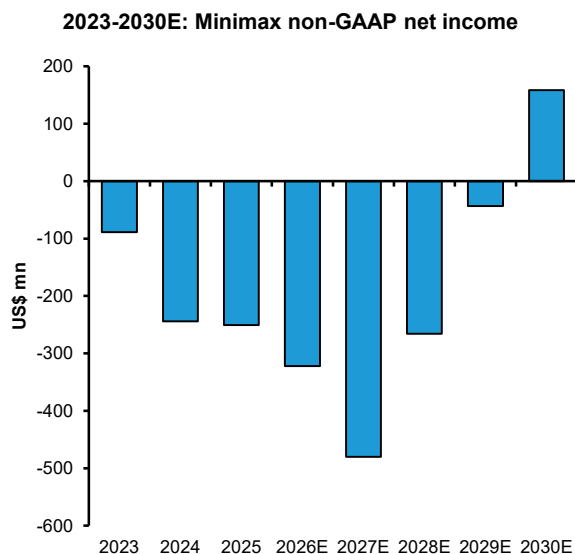
Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 69: **We model R&D spend rising to \$2.3bn by 2030**

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 70: **We model Minimax getting to operating profit breaking even in 2030**

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 71: **We model c. \$160mn non-GAAP net income in the same year**

Source: Corporate reports, Bernstein estimates and analysis.

VALUATION

Our HK\$275 price target for Minimax is based on a blend of (1) 4.5x 2029E PS, discounted back at 14% annually; and (2) our DCF for the company (see Exhibit 72). Minimax's shares will almost certainly rally sharply if M3 Pro lives up to its billing, but assuming success now feels hope rather than evidence based. Exhibit 73 shows our estimates for Minimax in the next few years, compared with Street consensus.

Exhibit 74 shows the pace at which Minimax's pre-IPO lock-ups expire, allowing its free float to rise. The company's lock-up expiry schedule points to its free float exceeding 70% by January 2027.

EXHIBIT 72: Minimax: DCF valuation

USD mn	2025	Est. 2026	Est. 2027	Est. 2028	Est. 2029	Est. 2030	Est. 2031	Est. 2032	Est. 2033	Est. 2034	Est. 2035
Revenue	79	419	1,005	2,217	3,807	5,554	7,164	8,454	9,468	10,131	10,435
Growth YoY		430.2%	139.8%	120.6%	71.7%	45.9%	29.0%	18.0%	12.0%	7.0%	3.0%
Gross profit	20	141	372	929	1,609	2,411	3,334	4,197	4,996	5,662	6,157
% of revenue	25.4%	33.7%	37.0%	41.9%	42.3%	43.4%	46.5%	49.7%	52.8%	55.9%	59.0%
R&D expenses	-253	-477	-764	-1,146	-1,662	-2,326	-2,902	-3,308	-3,575	-3,686	-3,652
% of revenue	-113.9%	-76.0%	-51.7%	-43.6%	-41.9%	-40.5%	-39.1%	-37.8%	-36.4%	-35.0%	-35.0%
Other operating expenses	-48	-114	-148	-181	-218	-259	-325	-372	-404	-419	-417
% of revenue	-27.2%	-14.7%	-8.2%	-5.7%	-4.7%	-4.5%	-4.4%	-4.3%	-4.1%	-4.1%	-4.0%
Profit/(loss) from operations	-281	-450	-540	-398	-271	-174	107	517	1,017	1,557	2,087
% of revenue	-355.6%	-107.4%	-53.7%	-18.0%	-7.1%	-3.1%	1.5%	6.1%	10.7%	15.4%	20.0%
Less: Income tax	0	0	0	0	0	0	-16	-78	-153	-234	-313
Effective tax rate %		15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%
NOPAT	-281	-450	-540	-398	-271	-174	91	440	865	1,324	1,774
Depreciation & amortization costs	1	4	11	24	45	70	95	117	134	147	154
Change in net working capital	6	34	83	182	313	456	589	695	778	833	858
Capital expenditure	-1	-6	-15	-33	-57	-83	-107	-127	-142	-152	-157
Additions to intangibles	0	-1	-3	-8	-13	-19	-24	-29	-32	-34	-35
Free cash flow	-275	-420	-465	-233	17	250	643	1,096	1,603	2,117	2,594
% of revenue	-348.2%	-100.2%	-46.3%	-10.5%	0.4%	4.5%	9.0%	13.0%	16.9%	20.9%	24.9%
WACC	14.4%										
Terminal growth rate	3.0%										
Discount factor		0.4	1.4	2.4	3.4	4.4	5.4	6.4	7.4	8.4	9.4
Present value of FCF - per share		-0.5	-1.1	-0.5	0.0	0.4	0.8	1.2	1.6	1.8	1.9
Sum of discounted cash flows - per share	6										
NPV of terminal value - per share	17										
Enterprise value per share (US\$)	23										
Less: minority interest - per share	0										
Add: adj. net cash/debt - per share	7										
Equity value per share (US\$)	31										
Equity value per share (HK\$)	239										
Implied 12-month forward price (HK\$)	273										
Upside/downside	18.9%										

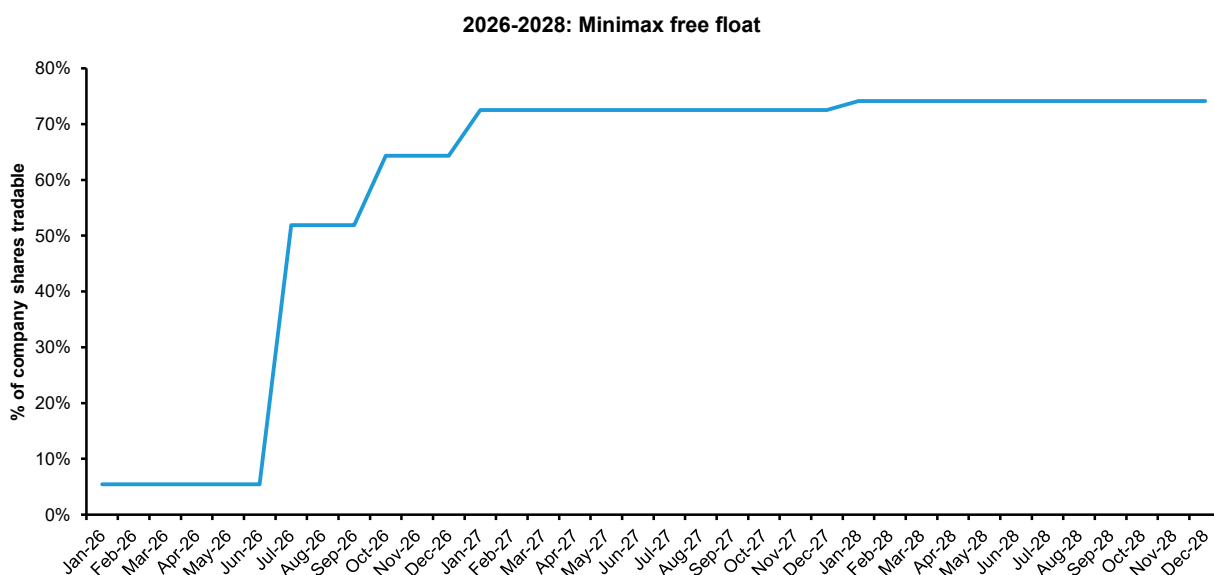
Source: Corporate reports and Bernstein analysis.

EXHIBIT 73: Minimax: Summary of estimates

Minimax: summary of estimates	2026E	2027E	2028E
Revenue			
Bernstein	419	1,005	2,217
Growth YoY	430.2%	139.8%	120.6%
Consensus	360	985	2,115
Growth YoY	403.7%	173.8%	114.8%
Bernstein vs. consensus	16.5%	2.1%	4.8%
GAAP operating income			
Bernstein	-450	-540	-398
Growth YoY	60.2%	19.9%	-26.2%
Margin	-107.4%	-53.7%	-18.0%
Consensus	-577	-635	-445
Growth YoY	66.5%	10.0%	-30.0%
Margin	-160.6%	-64.5%	-21.0%
Bernstein vs. consensus	-22.0%	-15.0%	-10.4%
Non-GAAP net income			
Bernstein	-322	-480	-266
Growth YoY	28.4%	49.1%	-44.6%
Margin	-76.8%	-47.8%	-12.0%
Consensus	-546	-574	-387
Growth YoY	79.9%	5.2%	-32.5%
Margin	-151.8%	-58.3%	-18.3%
Bernstein vs. consensus	-41.0%	-16.3%	-31.4%

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 74: Lockup expiry earlier this month drove a significant jump in the company's free float... there's more to go before year end



Source: Corporate reports and Bernstein analysis.

A RECAP ON TOKEN ECONOMICS, AND INFERENCE MARGINS

The following section is an abridged summary of our previous, longer discussion on AI model architectures and what they tell investors about the strategic choices made by their developers.

Bigger is usually better... but there are trade-offs

At a high level, frontier AI models have mainly scaled over time by adding (1) parameter count; (2) adding experts in sparse Mixture-of-Experts (MoE) architectures; and (3) post-training via techniques like supervised fine-tuning (SFT) and especially reinforcement learning (RL). In the last few years, the leading frontier models have increasingly gravitated towards sparse Mixture-of-Experts (MoE) architectures. These models have tended to have very high parameter counts, but a router only activates a subset of parameters (experts) at a time, based on specialisation on different vertical domains, languages, or skill sets. Compared with dense models where all of the parameters of the model are activated for all tokens, MoE models offer cheaper inference (inference cost correlates directionally with parameters activated, experts can be added without being activated for every token) and greater specialisation.

Looking across the top China AI labs, Minimax has generally had the smallest models, but also lower active parameter count per token. M2.5/2.7 for example has 230bn parameters but 10bn active at any given time. Minimax did not disclose the parameter scale of M3, but has hinted in private discussions that it's in the 400-500B range, with similar sparsity as the previous generation.

GLM-5.2 on the other hand has 744bn total parameters and 40bn active per token. At a high level, Minimax's smaller models prioritise efficiency and lower running costs... while maintaining competitive reasoning capabilities. Z.ai's much higher active parameter counts contribute to superior general reasoning, longer agentic task completion horizons, and lower hallucination rates... at the cost of higher inference costs per token.

The KV cache, and what pricing strategies tell us

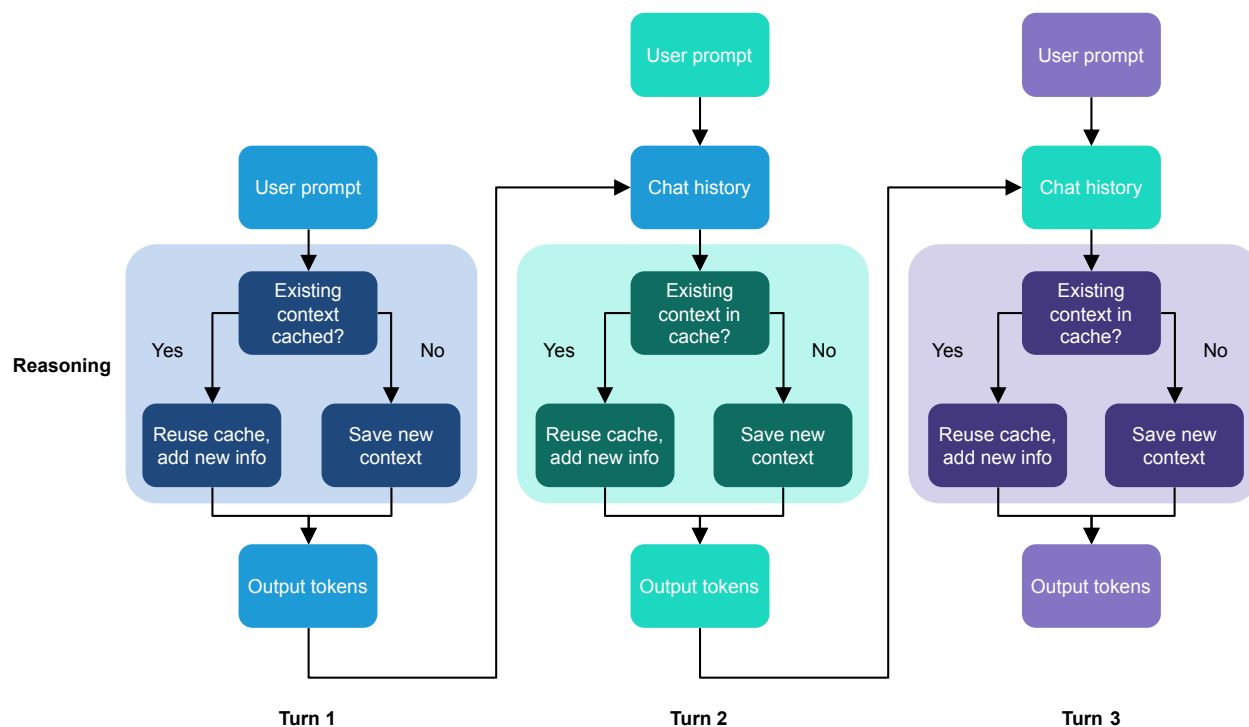
The Key Value (KV) cache is a form of short-term memory within AI models which stores tokens and instructions in preparation for repeated use during inference. Commonly stored data includes hidden instructions attached to the beginning of each user prompt (e.g. prefix caching), and vectors built based on the prior input context (such as conversation history in a thread). When a new prompt is inputted it's examined for matches with pre-existing KV segments so they can be re-used. In practice, for example, as users ask follow-up questions in a conversation thread, KV caching allows prior conversation to be reused without

needing to be re-calculated, while fresh tokens added on top are processed.

Exhibit 75 highlights a very basic schematic for how KV caching works in practice. Rather than go into fine technical detail on how KV caching works in practice, it's probably more useful to highlight that most models offer steep discounts for cache hit tokens, to reflect the fact said tokens only require decode and not pre-fill. Z.ai's GLM-5.2 model charges for cache hit tokens at a 75% discount to the standard input token price of RMB8/Mtok (c. \$1.40/Mtok). Exhibit 76 and Exhibit 77 show OpenRouter data showing cache hit rates for Minimax M3 and GLM-5.2.

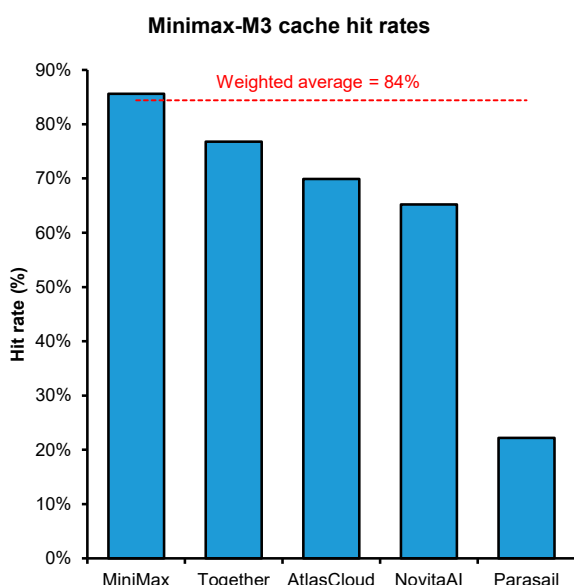
EXHIBIT 75: Architectural choices like parameter and active parameter scale, and use of the KV cache reflect the strategic priorities of the developer, and positioning along a cost-performance frontier

Indicative LLM reasoning and KV cache usage



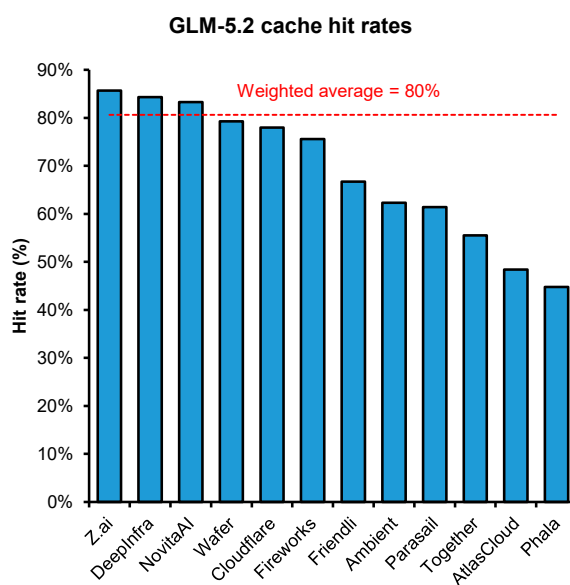
Source: Corporate reports and Bernstein analysis.

EXHIBIT 76: **The combination of agentic harness adoption and server level optimisation...**



Source: Openrouter and Bernstein analysis.

EXHIBIT 77: **...has driven cache hit rate improvements across the industry**



Source: Openrouter and Bernstein analysis.

PRICING POWER IN A COMPUTE CONSTRAINED MARKET

Aside from the progression of monthly ARR, we expect (1) blended token pricing; and (2) the progression of inference margins to represent the next most important variables to watch for Minimax and Z.ai. From a starting point early in the year when many investors considered the two companies comparable, our sense is a more nuanced understanding of the unit economics of these companies has started to emerge.

To illustrate, Exhibit 78 outlines our back of envelope model for AI inference revenue - and direct cost - per token. Exhibit 79 explains the key input variables for both. The architectural differences between Z.ai and Minimax's models mean they are exposed to different "typical" workloads which then drive differences in variables like input/output sequence length ratios, KV cache hit rates. On the margin, our bias is that Z.ai likely has a lower ISL:OSL ratio, reflecting the GLM models family's focus on general reasoning (which tends to produce longer outputs), a tendency for long reasoning traces, and on the other hand Minimax's models being more explicitly optimised for long-context agentic tool use.

To date, we believe that revenue/Mtok has been a much more impactful driver of Minimax and Z.ai inference margins to date than cost per token - the latter is more of a function of datacentre and GPU efficiency, where we think the companies should be comparable. Earlier this year Z.ai claimed an 83% increase in revenue per token year to date, while Minimax points to a 40% increase in blended rev/Mtok on account of higher priced cache hit tokens.

For Minimax though the fact half of its ARR has come from multimodal use cases strikes us as a risk factor, considering the workload mix shift year to date in favour of text/coding. Within each market the price for compute is likely to be somewhat similar (e.g. reflecting similar power and other infrastructure costs, the cost of GPUs, etc.).

EXHIBIT 78: **We model Z.ai and Minimax by referring to headline model pricing, third-party data points on cache hit rates, estimates for server level token through-put, and GPU rental costs**

$$Rev/Mtok = \left[\underbrace{\frac{IO}{(IO+1)}}_{\text{Input token mix}} \times \underbrace{P_{in}}_{\text{Input token cost}} \times \underbrace{((1-KV) + KV \times D_{kv})}_{\text{Discount for KV cache tokens}} + \underbrace{\frac{1}{(IO+1)}}_{\text{Output token mix}} \times \underbrace{P_{out}}_{\text{Output token cost}} \right] \times (1-C)$$

$$Cost/Mtok = \frac{Cost_{GPU-hr}}{\underbrace{(TPS \times U \times 3600/10^6)}_{\text{Tokens per GPU-hour}}}$$

Source: Bernstein analysis.

EXHIBIT 79: **Some simplification is unavoidable, but we're hopeful this helps us to rationalise what's going on under the bonnet at both labs**

Variable	Comments	
Rev/Mtok	Revenue per million tokens	
IO	Input vs. output sequence length ratio	- Ratio between input and output token lengths - Typically higher for coding and agentic tasks, shorter for general reasoning tasks with longer reasoning traces and end output
P-in	Input token price per million tokens	Listed input token price less commercial discounts
P-out	Output token price per million tokens	Listed output token price less commercial discounts
KV	KV cache hit rate	- Listed cache read token price less commercial discounts - KV cache hit ratios can vary considerably by workload, typically higher for repetitive agentic workloads, lower for general reasoning
D-kv	Cache read token discount vs. input token price	Listed output token price less commercial discounts, expressed as a percentage of non-cached input token price
C	Commercial discount	Reflects commercial strategy, timed promotions, customer mix, etc.
Cost/Mtok	Inference cost per million tokens	
T	Tokens per second per GPU	- Tokens calculated per second per GPU - Can be influenced by model architecture (e.g. parameter scale), batching, etc.
U	Utilisation	Assumed average GPU utilisation rate
Cost-GPU-hr	GPU cost per hour	Cost of GPU compute rental per hour, inclusive of overhead costs and hyperscaler margin if applicable

Source: Bernstein analysis.

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 80: Z.ai: Income Statement

RMB mn	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Revenue	125	312	724	5,631	18,663	44,435	85,401	146,813
Gross profit	80	176	297	1,522	5,855	15,250	34,601	70,622
Selling & marketing expenses	-101	-387	-391	-490	-722	-1,671	-2,899	-4,489
General & administrative expenses	-66	-134	-505	-714	-978	-2,089	-3,600	-5,331
Research and development expenses	-529	-2,195	-3,180	-5,089	-7,887	-12,226	-18,950	-29,372
Other items	-10	2	-7	0	0	0	0	0
Loss from operations	-626	-2,538	-3,787	-4,771	-3,732	-736	9,153	31,431
Non-GAAP operating profit	-459	-2,046	-2,251	-3,707	-2,612	1,930	14,277	40,240
Profit before tax	-788	-2,958	-4,718	-4,711	-3,702	-716	9,173	31,451
Income tax expense	0	0	0	0	0	-23	-417	-1,979
Net income	-788	-2,958	-4,718	-4,711	-3,702	-739	8,756	29,471
Net profit attributable to SHs	-445	-1,394	-1,918	-4,691	-3,685	-735	8,716	29,339
Non-GAAP net profit attributable to SHs	-278	-902	-382	-3,627	-2,565	1,931	13,841	38,148
Non-GAAP EPS per share - diluted (RMB)	-18.40	-5.64	-2.40	-7.95	-5.26	3.84	26.97	72.90

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 81: Z.ai: Cash Flow Statement

RMB mn	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Loss before taxation	-788	-2,958	-4,718	-4,711	-3,702	-716	9,173	31,451
Depreciation & amortization	69	280	281	209	330	839	2,035	4,208
Changes in working capital	-96	0	704	-1,743	-1,747	-3,712	-4,217	-5,279
Other	167	433	1,487	1,024	1,120	2,643	4,707	6,829
Cash flow from operations	-648	-2,245	-2,246	-5,220	-3,999	-945	11,698	37,208
Capex	-507	-126	-11	-563	-1,866	-4,444	-8,540	-14,681
Other	-278	77	-364	-56	-187	-444	-854	-1,468
Cash flow from investments	-785	-49	-376	-619	-2,053	-4,888	-9,394	-16,149
Cash flow from financing	2,463	3,312	2,614	18,760	25,000	15,000	10,000	10,000
Net effect of exchange rate changes	0	0	-5	0	0	0	0	0
Net change in cash	1,030	1,019	-13	12,920	18,948	9,167	12,304	31,059
Beginning cash	219	1,249	2,268	2,255	15,175	34,123	43,290	55,593
Ending cash	1,249	2,268	2,255	15,175	34,123	43,290	55,593	86,652

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 82: Z.ai: Balance Sheet

RMB mn	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Cash & cash equivalents, term deposits	1,249	2,269	2,368	15,290	34,241	43,412	55,723	86,793
Trade and other receivables	416	667	699	12,367	29,808	66,877	108,975	161,661
Other current assets	198	80	505	1,616	3,360	7,067	11,276	16,545
Total current assets	1,863	3,016	3,571	29,273	67,408	117,357	175,974	264,999
Property and equipment	788	866	656	1,022	2,582	6,251	12,916	23,722
Other non-current assets	209	494	627	670	833	1,213	1,908	3,043
Total assets	2,860	4,376	4,854	30,965	70,823	124,821	190,798	291,764
Trade and other payables	288	603	1,329	12,367	29,808	66,877	108,975	161,661
Other current liabilities	3,321	7,235	11,078	11,078	11,078	11,078	11,078	11,078
Total current liabilities	3,609	7,838	12,406	23,445	40,886	77,955	120,052	172,739
Other non-current liabilities	234	493	559	559	559	559	559	559
Total liabilities	3,843	8,331	12,965	24,003	41,444	78,514	120,611	173,298
Shareholders' equity	-983	-3,955	-8,111	6,962	29,379	46,307	70,187	118,467
Total liabilities & shareholders' equity	2,860	4,376	4,854	30,965	70,823	124,821	190,798	291,764

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 83: **Minimax: Income Statement**

USD mn	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Revenue	3	31	79	419	1,005	2,217	3,807	5,554
Gross profit	-1	4	20	141	372	929	1,609	2,411
Selling & marketing expenses	-23	-87	-52	-44	-60	-76	-92	-108
General & administrative expenses	-8	-14	-37	-70	-88	-105	-126	-151
Research and development expenses	-70	-189	-253	-477	-764	-1,146	-1,662	-2,326
Other items	9	36	40	0	0	0	0	0
Loss from operations	-92	-250	-281	-450	-540	-398	-271	-174
Profit before tax	-269	-465	-1,872	-451	-540	-399	-272	-175
Income tax expense	0	0	0	0	0	0	0	0
Net income	-269	-465	-1,872	-451	-540	-399	-272	-175
Net profit attributable to SHs	-269	-465	-1,872	-451	-540	-399	-272	-175
Non-GAAP net profit attributable to SHs	-89	-244	-251	-322	-480	-266	-43	158
Non-GAAP EPS per share - diluted (USD)	-0.85	-2.25	-2.31	-0.97	-1.36	-0.75	-0.12	0.44

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 84: **Minimax: Cash Flow Statement**

USD mn	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Loss before tax	-269	-465	-1,872	-451	-540	-399	-272	-175
Depreciation & amortization	1	2	3	2	7	15	30	53
Changes in working capital	26	14	6	165	262	470	663	658
Other	178	191	1,583	100	60	133	228	333
Cash flow from operations	-64	-258	-280	-183	-212	220	650	869
Capex	-1	-1	-1	-4	-10	-22	-38	-56
Other	-40	-431	73	0	0	0	0	0
Cash flow from investments	-40	-431	72	-4	-10	-22	-38	-56
Cash flow from financing	306	771	423	7,491	200	200	200	200
Net effect of exchange rate changes	0	1	3	0	0	0	0	0
Net change in cash	202	83	219	7,304	-22	397	812	1,014
Beginning cash	5	206	289	508	7,811	7,790	8,187	8,999
Ending cash	206	289	508	7,811	7,790	8,187	8,999	10,013

Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 85: **Minimax: Balance Sheet**

USD mn	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Cash & cash equivalents, term deposits	298	315	521	7,825	7,803	8,201	9,013	10,027
Trade receivables	1	7	11	46	110	242	417	609
Other current assets	20	483	475	475	475	475	475	475
Total current assets	320	806	1,007	8,346	8,389	8,918	9,905	11,111
Property and equipment	1	1	2	5	12	27	49	75
Other non-current assets	4	104	79	86	98	112	136	168
Total assets	324	911	1,088	8,438	8,499	9,057	10,090	11,354
Trade and bills payables	17	51	58	266	607	1,232	2,108	3,013
Other current liabilities	646	1,656	3,676	3,676	3,676	3,676	3,676	3,676
Total current liabilities	663	1,708	3,734	3,942	4,283	4,908	5,784	6,689
Other non-current liabilities	3	2	3	5,655	5,655	5,655	5,655	5,655
Total liabilities	666	1,710	3,737	9,598	9,939	10,563	11,439	12,345
Shareholders' equity	-342	-799	-2,648	-1,160	-1,440	-1,506	-1,349	-991
Total liabilities & shareholders' equity	324	911	1,088	8,438	8,499	9,057	10,090	11,354

Source: Corporate reports, Bernstein estimates and analysis.

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VALUATION METHODOLOGY**Z.AI Co., Ltd.**

We value Z.ai on a combination of 25x 2030E PE, discounted back at 14% annualised, and our DCF for the company.

MiniMax Group Inc.

We value Z.ai on a combination of 4.5x 2029E PS, discounted back at 14% annualised, and our DCF for the company.

RISKS**Z.AI Co., Ltd.**

The main fundamental downside risks related to Z.ai include the speed of enterprise AI adoption in China, and competition from rival AI model developers. Investor perceptions on the Chinese AI labs can vary significantly depending on the most recent model release, and can shift materially especially during periods of bearish AI sentiment.

MiniMax Group Inc.

The main upside risk related to Minimax's shares relates to a blockbuster M3 Pro release. The main fundamental downside risks related to Minimax include the speed of enterprise AI adoption in China, and competition from rival AI model developers.

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- **Underperform (UP):** The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

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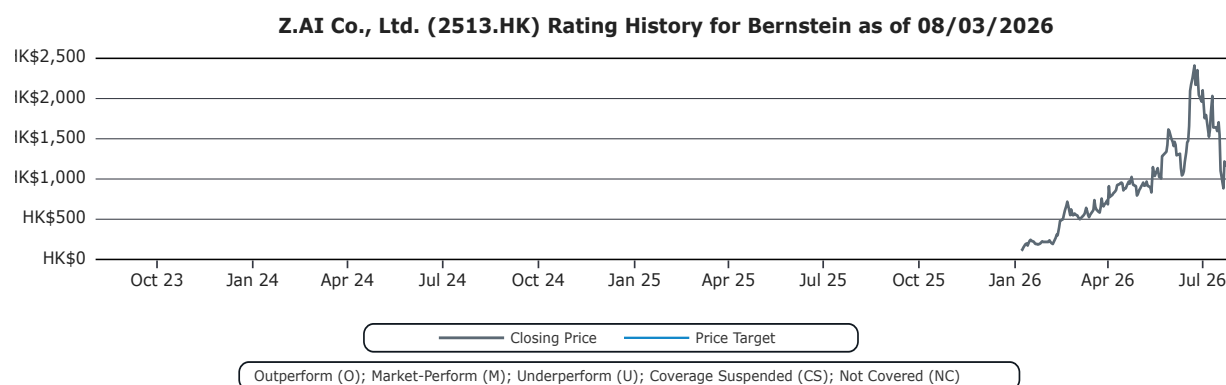
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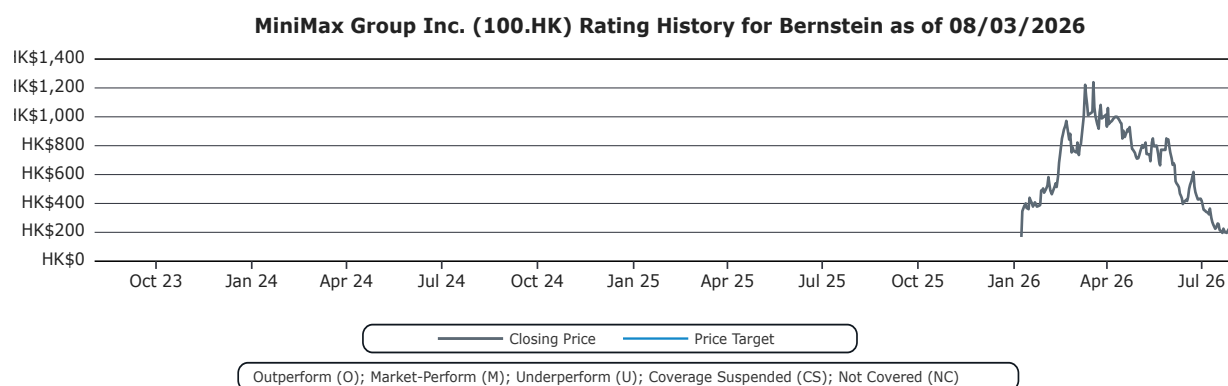
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